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SUBCRITICAL CRACK GROWTH IN GEOLOGICAL MATERIALS

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Abstract. A review is presented of the experimental data on subcritical crack growth in geological materials. The main parameters describing subcritical crack growth are the critical stress intensity factor K_c, the subcritical crack growth limit K_o, and the stress intensity factor-crack velocity (K-v) relationship between K_o and K_c. The K-v data are presented in terms of an equation in which the crack velocity depends on stress intensity factor raised to a power n because this is common practice in experimental studies. These data are presented as tables and in synoptic diagrams. For silicates the value of n increases as the environment becomes depleted in hyroxyl species and with increase in the microstructural complexity of the solid. Values of n as low as 9.5 have been found for tensile cracking of quartz in basic environments and as high as 170 for tensile cracking of basalt in moist air. Insufficient experimental data are available to predict subcritical crack growth behavior at depth in the earth's crust without major extrapolations of the data base. Schematic outlines are presented, therefore, of the probable influence on subcritical crack growth of some key parameters in the crustal environment. These include stress intensity factor, temperature, pressure, activity of corrosive environmental agent, microstructure, and residual strains. In addition, a discussion is presented of the likely magnitude of the subcritical crack growth limit. For stress corrosion tensile crack growth of quartz a limit of approximately 0.2 of the critical stress intensity factor is inferred from theoretical calculations. Further problems discussed with regard to the extrapolation of experimental data to crustal conditions include the choice of a suitable equation to describe crack growth and the magnitude of parameters in these equations. A brief discussion of the double torsion testing method is presented in order to aid the interpretation of experimental results because it is almost the sole method used to study subcritical cracking in rocks.

Introduction

In a linear elastic solid an isolated sharp slit, plane crack, whose walls are free of traction, will propagate at speeds approaching that of sound in the medium once a critical stress intensity factor K_c has been exceeded at the crack tip. There are three fundamental modes of crack tip displacement: tension (mode I), in-plane shear (mode II), and antiplane shear (mode III). The near-field stress distribution for any mode is given in terms of the stress intensity factor by [Lawn and Wilshaw, 1975]

$$\sigma_{ij} = K_L(2\pi r)^{-1/2} f_{ij}(\theta) \quad (1)$$

where $f_{ij}(\theta)$ is a well-defined function of θ, the angle measured from the plane ahead of the crack, and depends upon the loading mode, K_L is the stress intensity factor which embodies the loading configuration (the subscript L denotes the mode of loading: I, II, or III) and essential boundary conditions of the crack system, and r is the distance from the crack tip. For a two-dimensional crack in any mode the stress intensity factor is given by

$$K = Y \sigma_r (\pi c)^{1/2} \quad (2)$$

where σ_r is the remote applied stress, Y is a numerical modification factor to account for crack geometry, loading conditions, and edge effects, and c for penny-shaped, internal cracks is half the crack length. A vast literature on the fracture of metals, glasses, ceramics, and rocks attests to the utility of approaching problems involving catastrophic crack propagation through the critical stress intensity factor. This is not, however, a sufficiently general view of crack growth during long-term loading. In experiments it is found that significant rates of crack extension can occur at values of stress intensity factor that can be substantially below K_c. This phenomenon is known as subcritical crack growth, stable cracking, or quasi-static crack propagation. It can be due to several competing mechanisms, some of which are dependent upon the chemical environment. The most familiar of these is stress corrosion. Stress corrosion involves the weakening of strained bonds at crack tips by the chemical action of an environmental agent, such as water, thereby facilitating crack growth. Subcritical crack growth occurs at velocities which decrease as K is lowered to some threshold value K_o below which no crack propagation is observed. The details of the crack growth mechanism and the interaction between the environment and the microstructure of the solid control the details of the stress intensity factor/crack velocity curve between K_c and K_o. A schematic stress intensity factor/crack velocity diagram for subcritical tensile crack growth by stress corrosion is shown in Figure 1. This figure shows a stress corrosion threshold, although it is not yet clear whether in fact all materials behave in this way [Freiman, this issue]. A stress corrosion limit has been observed in soda-lime silicate glass and borosilicate glasses [see Freiman, this issue], but these observations have not been extended to other glasses nor to ceramics or rocks.

Figure 1 shows the now classical trimodal behavior familiar from work on glass. In region 1 the velocity of crack growth is controlled by

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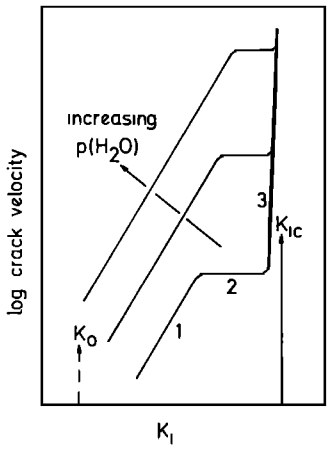


Fig. 1. Schematic stress intensity factor/crack velocity diagram for subcritical tensile crack growth due to stress corrosion. K_{Ic} is the critical stress intensity factor or fracture toughness, K_o is the stress corrosion crack growth limit. Region 1 behavior is controlled by the rate of stress corrosion reactions at crack tips. Region 2 behavior is controlled by the rate of transport of reactive species to crack tips. Region 3 behavior is controlled by thermally activated bond rupture which is largely independent of chemical environment.

the rate of stress corrosion reactions at crack tips. In region 2 the velocity of crack growth is controlled by the rate of transport of reactive species to crack tips. In region 3, crack growth is mainly controlled by mechanical rupture and is relatively insensitive to the chemical environment [Freiman, this issue]. It is believed that, in principle, the above statements hold for all three fundamental modes of crack tip displacement, although there is little experimental evidence in support of this.

This paper consists of two major parts. First, a comprehensive survey of the available experimental data on subcritical crack growth in geological materials is presented. Second, the available data are extrapolated to conditions of interest in the earth's crust. For variables whose influence on subcritical cracking of rocks has not been studied experimentally, schematics of probable trends are presented based upon work with ceramics and upon theoretical perspectives.

The vast majority of experimental studies of subcritical cracking in geological materials have been made using the double torsion testing method. Because of this a discussion of the method is presented in the appendix.

Theoretical Background

Crack Driving Force

Instead of considering crack tip stresses, as in equation (1), an alternative fracture mechanics approach to crack extension is to consider the strain energy release rate G. This is a global energy criterion rather than a local fracture criterion. Catastrophic crack extension occurs when a critical value G_c has been exceeded. The value of G includes contributions from all crack tip displacement modes. For plane strain, G is given by

$$G = K_I^2(1-p^2)/E + K_{II}^2(1-p^2)/E + K_{III}^2(1+p)/E \quad (3)$$

where the subscripts to the stress intensity factors identify the different modes of crack tip displacement, p is Poisson's ratio, and E is Young's modulus. It can be seen that the strain energy release rates for different modes are additive.

Considering, as an example, purely tensile loading, G is related to the Griffith energy balance concept by

$$G_c = K_{Ic}^2(1-p^2)/E = 2Q_f \quad (4)$$

where Q_f is the tensile fracture surface energy or global fracture resistance term. The fracture surface energy is generally greater than the ideal or thermodynamic surface energy Q_s, which in ideally brittle solids is merely the work done in separating material across neighboring atomic planes. This is because various energy dissipation processes operate at crack tips in many solids, other than those processes associated purely with new surface formation. These additional processes include crack tip microplasticity and acoustic emission, and they are all included in the calculation of the fracture surface energy.

It should be noted here that it is assumed throughout this paper that rocks and minerals behave according to linear elastic fracture mechanics. The stress/strain behavior of some rocks is known to be nonlinear, however, and the use of linear elastic parameters may be questionable [Schmidt and Lutz, 1979]. It has been shown, however, that at room temperature the values of linear elastic fracture parameters for Westerly granite and Nevada tuff are consistent with J integral values, that is, a fracture mechanics parameter that does not assume linear elastic behavior [Schmidt and Lutz, 1979; Weisinger et al., 1980]. One might expect nonlinear behavior to be enhanced by raising the temperature. For quartz, however, there is no nonlinear zone of plastic deformation at crack tips for temperatures up to 250°C (see section on mechanisms of crack growth).

Subcritical Crack Growth

A number of equations have been proposed to describe subcritical crack growth. Some of these are very complex because attempts were made to provide as comprehensive a description of the competing mechanisms as possible. Reviews of relevant theoretical studies are given by Anderson and Grew [1977] and by Atkinson [1982]. Few of these equations have found much favor with experimentalists, largely because of the complexity of the resulting descriptions of crack growth and the difficulty involved in measuring the parameters in these descriptions.

The most popular equation used to describe subcritical crack growth in geological materials is the well-known Charles [1958] power law:

$$v = v_o \exp(-H/RT)\, K_I^n \quad (5)$$

where v is the crack velocity, H is the

activation enthalpy, R is the gas constant, T is the absolute temperature, and v_o and n are constants. Sometimes, n is known as the stress corrosion or subcritical crack growth index.

Another popular equation is the Wiederhorn and Bolz [1970] modification and simplification of the Charles and Hillig [1962] theory of stress corrosion. Crack velocity is given by

$$v = v_o' \exp[-H + (2V^*K_I)/(\pi r_c)^{1/2}]/RT \quad (6)$$

where v_o' is a constant, V^* is an activation volume, and r_c is the radius of curvature of the crack tip. This equation is further simplified for empirical fits to experimental data to

$$v = v_o' \exp(-H + bK_I)/RT \quad (7)$$

where b is an experimentally determined constant and equation (6) is identical with equation (7) when $V^* = (b/2)(\pi r_c)^{1/2}$.

Equation (6) can only be used to describe region 1 of Figure 1, whereas equation (5) can be used to describe the whole of the figure with appropriate changes in the empirical constants [Atkinson, 1982].

In practice, because of the strong dependence of crack velocity on stress intensity factor it is almost impossible to distinguish between equations (5) and (7) for most data sets. This would be no problem were it not for the need to extrapolate experimental data in order to make predictions concerning tectonic deformation. More of this point will be made later.

Mechanisms of Subcritical Crack Growth

We can suggest several possible mechanisms of environmentally dependent subcritical crack growth. These include stress corrosion, dissolution, diffusion, ion exchange, and microplasticity. All of these are likely to be influenced by the chemical effects of pore water in crustal rocks.

Studies of crack propagation in glass under vacuum have shown that subcritical cracking is possible even in the absence of chemical agents [see Wiederhorn, 1974]. Various theories have been proposed to explain this behavior, based up on the concept that crack propagation is controlled by the atomic configuration of the crack tip [e.g., Fuller and Thomson, 1978; Lawn, 1975]. These "atomistic" theories assume that for a crack placed at the center of an array of atoms there is a stress intensity for crack growth and a lower one for crack healing. Between these two there exists a range of stress intensities for which the crack is "lattice trapped." Thermal activation over the barrier or quantum mechanical tunneling through the barrier can allow the crack to grow to a new position of stability. In Lawn's [1975] atomistic theory a lattice-trapped crack grows by lateral motion of atomic kinks along the crack front. This theory deals with intrinsic bond rupture processes. There is nothing, however, to exclude the incorporation of extrinsic processes, such as the action of water vapor. The role of an external agent may be viewed in terms of an enhanced activation.

The physicochemical details of the mechanisms of subcritical cracking in rocks, and in other materials, are rather poorly understood. Before much more progress can be made in extrapolating laboratory experimental data to conditions in the earth's crust it will be necessary to have a much better understanding of these details.

Stress Corrosion

It is assumed in much of this paper that stress corrosion is the main mechanism of subcritical crack growth in shallow crustal conditions (upper 20 km). Fracture mechanism maps for quartz [Atkinson, 1982] lend support to this assertion by suggesting that the propagation of preexisting cracks will be the dominant mode of failure at these crustal levels.

For silicates it is envisaged that strained Si-O bonds at crack tips can react more readily with environmental agents than unstrained bonds because of a strain-induced reduction in the overlap of atomic orbitals [Michalske and Freiman, 1982]. The reaction between strained bonds and environmental agent produces a weakened (activated) state which can then be broken at lower stresses than the unweakened bonds to yield some product. This reaction can be expressed as

$$XA + B \leftrightarrows B^* = \text{products} \quad (8)$$

where X is the number of environmental species A reacting with a single crack tip bond B to give the weakened, or activated, state B*. For silicate glasses and quartz in water environments a general expression for weakening has been proposed [Scholz, 1972; Martin, 1972; Swain et al., 1973; Atkinson, 1979b] as follows:

$$H\text{-}O\text{-}H + [\equiv Si\text{-}O\text{-}Si\equiv] \leftrightarrows \equiv Si\text{-}OH.HO\text{-}Si\equiv \leftrightarrows 2[\equiv Si\text{-}OH] \quad (9)$$

The hydrolysis of strong Si-O bonds to weaker hydrogen bonded hydroxyl groups linking the silicon atoms as shown above is a first-order chemical reaction, that is, X = 1 in equation (8). Although many writers assume that equation (9) is the key stress corrosion reaction in silicate minerals, this assumption is largely based on the results of experimental work done on glasses.

In basic environments it has been proposed that silica glasses undergo corrosion according to [Charles, 1958]

$$\equiv Si\text{-}O\text{-}Si\equiv + OH^- \leftrightarrows \equiv Si\text{-}O^- + \equiv Si\text{-}OH \quad (10)$$

Stress corrosion of Si-O-Si bonds has been attributed to both ionized water and to molecular water [Michalske and Freiman, 1982]. Freiman [this issue] shows that it is the activity of the corrosive agent that is of importance in controlling the rate of stress corrosion. The activity a_i of a species under fixed conditions is given by the ratio of the fugacity under those conditions to that in a convenient reference state.

Michalske and Freiman [1982] believe that they can account for stress corrosion of glass in terms of a molecular interaction between water

TABLE 1. Calculated Tensile Crack Growth Rates in Quartzite Controlled by Grain Boundary Diffusion

Grain Size, μm	Temperature, °C	K_I, MPa $m^{1/2}$	Crack Velocity, m s^{-1}
10	200	1	7×10^{-13}
10	200	0.5	6×10^{-14}
10	400	1	2×10^{-10}
10	400	0.5	2×10^{-11}
1000	200	1	7×10^{-15}
1000	200	0.5	5×10^{-16}
1000	400	1	2×10^{-12}
1000	400	0.5	2×10^{-13}

and Si-O bonds. It is found, however, that changing the activity of hydroxyl ions can significantly alter the rate of crack propagation in quartz in aqueous environments [Atkinson and Meredith, 1981]. Increasing hydroxyl ion concentration results in faster crack velocities, but the activity of molecular water is essentially constant during these tests. Ionized water would seem to be an important agent in the stress corrosion process. It is of some geophysical significance to determine which are the important species for stress corrosion in the earth's crust, and this problem will be addressed later.

For calcite rocks the nature of stress corrosion reactions is even less well understood. It may well involve reactions such as those described in the section below on dissolution.

More complex, and perhaps more physically realistic, "water" incorporation reactions are possible in quartz and other silicates [see Dennis, this issue]. Unfortunately, they are not well enough understood to be related to the present work. Crustal rocks are commonly multiphase aggregates and not the monomineralic idealized test materials often used by experimenters. It is not clear to what extent any of the proposed stress corrosion weakening reactions apply to more complex rock-forming minerals than quartz, such as feldspar, mica, pyroxenes, amphiboles, and olivines.

It is generally observed that as crack velocity decreases, the crack path tends to become predominantly intergranular. Thus at geophysically interesting slow crack velocities, stress corrosion will be controlled by the details of grain boundary surface chemistry and the reaction of grain boundaries and matrix material with the stress corrosion agent. We have virtually no relevant data here, but an understanding of this phenomenon is also required for reliable extrapolation of laboratory data to geophysical problems.

Diffusion

Although mass transport processes are unlikely to be important in subcritical crack propagation in the uppermost crustal levels except at very slow rates of strain (and crack velocities), it is of interest to estimate how important it might be at hotter and/or deeper crustal levels. Experimental studies of ceramics have shown that crack growth at high homologous temperatures due to mass transport can be the dominant mechanism of subcritical crack growth in deformation at engineering rates of strain.

The problem has been analyzed theoretically by Stevens and Dutton [1971] and by Dutton [1974]. A number of potential diffusion paths are possible, including lattice (bulk) diffusion, surface diffusion, vapor phase diffusion, and grain boundary diffusion. For lack of suitable data, surface and vapor phase transport are not considered here. Grain boundary diffusion rates are likely to be faster than volume diffusion rates at relatively low homologous temperatures. Tensile crack growth by grain boundary diffusion of vacancies to crack tips is given by [Dutton, 1974]

$$dc/dt = [V_a D_b w/4r_c^2 dkT][B\sigma^2 c - Q_b + \sigma r_c] \quad (11)$$

where V_a is the "atomic" volume, D_b is the grain boundary diffusion coefficient, w is the grain boundary width, r_c is the crack tip radius, d is approximately half the grain diameter, σ is the remote applied tensile stress, c is half the crack length, Q_b is the specific grain boundary surface energy, and $B = \pi(1-p^2)/2E$. The value of stress intensity factor can be calculated from equation (2).

Assuming that for quartzite the diffusion of oxygen is the deformation rate-controlling step, then the arrival of one oxygen atom at a lattice site is accompanied, on average, by the arrival of 0.5 of a silicon atom. Thus V_a is half the molecular volume of quartz, that is, $V_a = 1.9 \times 10^{-29}$ m^3. Also $w = 10^{-7}$ m, Young's modulus E at temperature T is given by $E_T = E_o(1-XT)$ where E at 0°K, E_o, = 103×10^9 Pa and $X = 2.17 \times 10^{-4}$ $°K^{-1}$, $D_b = 2.7 \times 10^{-21}$ m^2 s^{-1} (200°C), 1.2×10^{-18} $m^{2b} s^{-1}$ (400°C), $Q_b = 1$ J m^{-2} [see Dennis and Atkinson, 1984; Dennis, this issue] and $r_c = 5 \times 10^{-10}$ m [Spetzler et al., 1981]. We can thus calculate the rate of crack growth for given values of K_I as shown in Table 1.

It can be seen that although the calculated velocities are slow, they are not trivial on a geophysical time scale. Although space does not permit diffusion-controlled subcritical crack growth to be examined further here, these rough calculations suggest that it merits further study. At temperatures typical of the upper crust, however, it is likely that diffusion-controlled subcritical crack growth will be important only at relatively low stress intensity factors. At relatively high values of stress intensity factor, other mechanisms, such as stress corrosion, will be more important (see later).

Some solids contain dissolved chemical impurities, such as structurally bound water in quartz, which can have a degrading effect on plastic flow stress, even when present in remarkably small concentrations (0.001 to 0.100 weight %) [e.g., Griggs and Blacic, 1965; Hobbs et al., 1972]. During crack propagation, stress-directed diffusion of these chemical impurities to crack tips may occur, where they

may be able to take part in weakening reactions, thus facilitating crack extension [Dutton, 1974]. For silicates, stress-directed diffusion of structurally bound water to crack tips could facilitate stress corrosion reactions or dislocation activity (see section on microplasticity), both of which could allow subcritical crack growth.

Dissolution

It has been found for binary alkali-silicate glasses tested in aqueous solutions that there is a velocity plateau in tensile stress intensity factor/crack velocity diagrams at low crack velocities of 10^{-8} to 10^{-10} m s^{-1} [Simmons and Freiman, 1981]. It is suggested by Simmons and Freiman [1981] that because of the great solubility of these glasses in water the plateau results from cracks growing at a rate controlled by the silica dissolution rate. A very small stress is apparently all that is required to remove corrosion products from the crack tip and accelerate dissolution there rather than at the sides of the crack. As will be seen later it is possible that this phenomenon has been observed in calcite rocks. In principle, there is no reason why dissolution should not be important in crack growth for other materials at low stress intensity factors, provided that the dissolution rate is fast enough.

The solubility of calcite is quite high at room temperature and is much higher than that of quartz. Solution of calcite is controlled by the net equation [Fyfe et al., 1978]

$$CO_2 + H_2O + CaCO_3 \leftrightarrows Ca^{2+} + 2HCO_3^- \qquad (12)$$

Solubility increases with rising partial pressure of CO_2 and decreasing temperature. Solubility is greater in NaCl solutions and seawater than in fresh water. This retrograde temperature dependence of solubility is in marked contrast to the behavior of quartz which shows an increase in solubility with increase in temperature (normal solubility) at all moderate pressures [Fyfe et al. 1978]. Solubility of quartz is largely unaffected by dissolved salts. Solubility is also largely independent of pH, except at pH>9, where there is a large increase in solubility as pH is raised. At high temperatures and pressures the solubility of quartz is slightly decreased in chloride solutions and increased in alkaline solutions. It is commonly assumed that the silica species in solution is neutral, either $SiO_2.2H_2O$ or $Si(OH)_4$ in view of the lack of influence of changing pH. The hydration number x may vary, however, with temperature, solution density, and pH. In the aqueous dissolution of quartz through the reaction

$$ySiO_2 + xH_2O \leftrightarrows ySiO_2.xH_2O_{aq} \qquad (13)$$

the values of y and x are usually taken as 1 and 2, giving $Si(OH)_4$ in solution. If the polymeric species are neutral, then silica solubility will be unaffected by increased ionic strength or varying pH, but a small amount of acid dissociation must occur.

Ion Exchange

If the chemical environment contains species which can undergo ion exchange with species in the solid phase and if there is a gross mismatch in the size of these different species, then lattice strain can result from ion exchange which can facilitate crack extension, for example, exchange of H^+ for Na^+ in silicate glasses [Wiederhorn, 1978]. For modified E-glass fibers in water the replacement of alkali ions by smaller hydrogen ions produces a tensile stress at the glass surface that can be high enough to result in spontaneous fracture [Wiederhorn, 1978].

Another effect of ion exchange is to modify the chemistry of the crack tip solution. For glass/water systems the exchange of hydrogen ions for alkali ions, which occurs rapidly even at room temperature, produces a basic solution at the crack tip. The pH can quickly reach high values because of the restricted volume of fluid at the crack tip. Once pH exceeds 9, direct attack of the silicate network occurs [Wiederhorn, 1978]. The importance of these ion exchange reactions for subcritical crack propagation will depend on how easily the crack tip solution is modified by diffusional exchange with the bulk external environment.

A two-stage pattern of behavior is envisaged, the details of which are controlled by the crack velocity, the rate of reaction between the external environment and the material of the crack tip, and the chemistry of the solid in the vicinity of the crack tip [Wiederhorn and Johnson, 1973]. At high crack velocities, crack growth is controlled by the chemical composition of the new crack surfaces. Transport of chemical species from the bulk fluid to the crack tip environment cannot keep pace with creation of new sources of reactive ions in fresh crack surfaces. At low crack velocities, crack growth is controlled by the chemistry of the external environment. This is because at these low crack velocities the crack tip environment is open to modification by the external environment through diffusion of chemical species along the crack. Chemical differences between crack tip fluid and the bulk fluid are not sustained for any significant length of time because crack velocity is very slow. Almost as soon as fresh crack tip surface is created it becomes depleted in reactive ions through the reactions with the bulk fluid [Wiederhorn and Johnson, 1973; Wiederhorn, 1978]. See Atkinson and Meredith [1981] and later sections for a discussion of the similar behavior of quartz.

Microplasticity

Several studies have suggested that plastic flow may be a viable mechanism of crack propagation in glass, even at room temperature [e.g., Marsh, 1964; Wiedmann and Holloway, 1974; Williams and Marshall, 1975]. It has yet to be shown, however, that these plastic flow models agree in detail with the huge body of empirical data on the environmental dependence of crack velocity in glass.

Dislocation mobility in silicates is thought to be strongly influenced by the presence of

structurally bound water [e.g., Hobbs et al., 1972]. The concentration of structurally bound water in silicates is dependent upon the chemical and physical environment. Microplasticity at crack tips in silicates may be an important mechanism of crack extension [Gandhi and Ashby, 1979; Atkinson, 1982] and could lead to subcritical crack growth, provided that the homologous temperature and/or structurally bound water concentration were high enough.

Dislocation activity allowing plastic flow may be of significance even in low-temperature subcritical cracking of some minerals, such as galena or calcite. All of the available evidence from electron microscopy of quartz, however, suggests that chemically enhanced subcritical crack growth is not accompanied by any significant plastic flow at crack tips, at least at temperatures up to 250°C [Martin and Durham, 1975; Dunning et al., 1980; Lawn et al., 1980].

Factors Influencing Subcritical Crack Growth

The factors influencing subcritical crack growth will depend upon the particular mechanisms involved. In general, however, the following variables will be important: (1) stress intensity factor (K), (2) crack tip displacement mode, (3) temperature, (4) pressure (total overburden pressure P_t, pore fluid pressure P_f, and effective pressure ($P_e = P_t - P_f$)), (5) activity of corrosive environmental agents and pH, (6) solubility of solid in environmental agents, (7) microstructural variables: mineralogy, crystallography, grain size/morphology/distribution, porosity, fabric, degree of heterogeneity, microcrack density/length/distribution; the influence of microstructural variables is felt mainly through their effect on the process zone size and configuration, that is, the nonlinear microcrack zone at macrocrack tips, and (8) residual strain of palaeotectonic origin.

It is also important to consider the testing method used to obtain stress intensity factor/crack velocity data. As will be seen later, rather different K-v data are obtained by different workers using the same testing method. The double torsion test is the main method used on rocks, yet it is prone to some significant problems (see the appendix and Swanson [this issue]). These points must be borne in mind when evaluating the significance of any set of K-v data.

In this paper, crack growth is discussed in terms of the stress intensity factor attained at the high-velocity limit to quasi-static or subcritical crack growth K_c, the stress intensity factor attained at the lower crack velocity limit to subcritical crack growth K_0, and the slope and other characteristics of the K-v curve between these end points. The slope of the K-v curve is expressed in experimental results by the n of equation (5) and the b of equation (7). Subcritical crack growth data are discussed here in terms of K_c, K_0, and n because this corresponds to the most common experimental usage.

Excepting the studies of Cox and Atkinson [1983] and Okubo and Dieterich [1981], there are virtually no experimental fracture mechanics data on single macroscopic shear cracks in rocks. The shear fracture energies derived from the triaxial experiments of Wong [1982] include contributions from several sources. The final failure plane is merely the most favored of many potential surfaces, all of which may have had work done on them in the build up to catastrophic failure. This review will concentrate, therefore, on experimental data for the tensile crack tip displacement mode.

Experimental Data for Tensile Crack Propagation

Compilations of the experimentally determined values for K_{Ic} and the subcritical crack growth index n for rocks and minerals are presented in Tables 2, 3, and 4. Data for the parameter v_o are not given because of the considerable uncertainty in these values when determined by the relaxation form of double torsion testing (see the appendix and Swanson [this issue]) and because it is often difficult to extract data on v_o from published work. In experiments it is not always the parameter K_{Ic} that is determined. Some workers choose G_{Ic} or Q_f, the critical strain energy release rate or the fracture surface energy, respectively, as the parameter to measure. Consequently, these other parameters are also quoted in Tables 2 and 3 in cases where they have been determined. Calculated values of the equivalent parameters not measured in the experiments are also presented in cases where the values of the elastic constants are known. To make comparisons between fracture surface energy Q_f and the thermodynamic surface energy Q_s calculated values of the thermodynamic surface energy derived from molecular theory are presented in Table 5. The temperature dependence of subcritical crack growth is indicated in Table 4 by the experimentally determined values of the activation enthalpy.

After presenting the experimental data for geological materials they are discussed in terms of the factors outlined in the previous section that are known to influence subcritical crack growth. For some of the rock types considered here Ouchterlony [1980] has presented a compilation of fracture mechanics parameters for catastrophic tensile failure which gives more detailed experimental data.

Quartz

Figure 2 shows the influence of both stress intensity and crystallography on subcritical crack growth. The well-known three-stage behavior, familiar from work on glasses and ceramics and reviewed in the introduction (see Figure 1), is clearly evident. There is a substantial anisotropy in subcritical crack growth behavior arising from crystallographic anisotropy in the mineral (see also Tables 3 and 4).

Estimates of the critical stress intensity factor for natural Brazilian quartz are much higher than those for synthetic quartz (see Norton and Atkinson [1981] and Table 3). There may be important differences in the fracture behavior of natural and synthetic quartz, but this area remains to be explored in systematic experiments.

TABLE 2. Experimental Values of Mode I Fracture Mechanics Parameters and Calculated Equivalents for Rocks

Material	K_{Ic}, MPa $m^{1/2}$	G_{Ic}, J m^{-2}	Q_f, J m^{-2}	Method	Reference
Quartz Rocks					
Tennessee sst	(1.43)	(76)	38	3PB	Friedman et al.[1972]
	0.45	(7.2)	(3.6)	DT	Atkinson[1979a]
	(2.24)	(176)	88	DCB	Perkins and Krech[1966]
Ruhr sst	1.43	(55.4)	27.7	3PB	Winter and Rummel[1979]
Berea sst	0.28	(113)	(57)	HF	Zoback[1978]
Bedding plane, parallel to bedding	(0.57-0.61)	465-520	(233-260)	DCB	Hoagland et al.[1973]
Normal to bedding plane, parallel to bedding	(0.65-0.73)	600-750	(300-375)	DCB	Hoagland et al.[1973]
Normal to bedding plane, normal to bedding	(0.89-1.06)	1120-1580	(560-790)	DCB	Hoagland et al.[1973]
Bedding plane, parallel to bedding	(0.33-0.44)	150-270	(75-135)	DCB	Hoagland et al.[1973]
Normal to bedding, parallel to bedding, 196°C	(0.46)	300	150	DCB	Hoagland et al.[1973]
Coconino sst	(0.65)	(50)	25	3PB	Friedman et al.[1972]
	(1.46)	(254)	127	DCB	Perkins and Bartlett[1963]
Milsap sst	(1.55)	(228)	114	DCB	Perkins and Bartlett[1963]
Colorado sst	(2.53)	(192)	96	DCB	Perkins and Barlett[1963]
Woodbine sst	(0.06)	(19)	9.5	DCB	Perkins and Bartlett[1963]
Torpedo sst	(1.0)	(122)	61	DCB	Perkins and Bartlett[1963]
Boise sst	(0.67)	(88)	44	DCB	Perkins and Bartlett[1963]
Coarse grain porous sst. 6.8% porosity	1.46			IPTWC	Clifton et al.[1976]
7.9% porosity	0.57			IPTWC	Clifton et al.[1976]
8% porosity	0.65			IPTWC	Clifton et al.[1976]
13% porosity	0.81			IPTWC	Clifton et al.[1976]
"sandstone"	0.31-0.35			3PB	Suzuki et al.[1978]
Chilhowee qtzt			31-62	3PB	Friedman et al.[1972]
Arkansas novaculite	1.34	(24)	(12)	DT	Atkinson[1980]
Mojave qtzt	2.1	(132)	(66)	DT	this study
Sioux qtzt	(2.45-2.65)	180-210	(90-105)	WDCB	Peck and Gordon[1982]
	(2.4-2.47)	173-182	(87-91)	WDCB	Peck and Gordon[1982]
Calcite Rocks					
Solnhofen lst	0.78	(10.2)	5.1	3PB	Winter and Rummel[1979]
Normal to bedding	1.01	(16.7)	(8.4)	DT	this study
Parallel to bedding	0.87	(12.5)	(6.3)	DT	this study

TABLE 2. (continued)

Material	K_{Ic}, MPa $m^{1/2}$	G_{Ic}, J m^{-2}	Q_f, J m^{-2}	Method	Reference
Indiana (Salem)	(0.75-	(32-	16-	3PB	Friedman et al.[1972]
1st	0.88)	44)	22		
Bedding plane,	(0.94-	50-	(25-	DCB	Hoagland et al.[1973]
parallel to	1.15)	75	38)		
bedding					
Normal to bedding	(1.85-	195-	(98-	DCB	Hoagland et al.[1973]
plane, parallel	2.01)	230	115)		
to bedding					
Normal to bedding	(1.25-	90-	(45-	DCB	Hoagland et al.[1973]
plane, normal	1.39)	110	55)		
to bedding	(1.22)	85	(43)	DCB	Hoagland et al.[1973]
Normal to bedding	(0.84)	40	20	DCB	Hoagland et al.[1973]
plane, normal					
to bedding,					
wet, 196°C					
Indiana 1st	0.99	(56)	(28)	3PB	Schmidt[1976a]
	0.69-	(27-	(14-	3PB	Schmidt[1975]
	0.99	56)	28)		
	(1.22)	(84)	42	DCB	Perkins and Bartlett[1963]
Indiana 1st					
pressure, MPa					
0	0.93	(49)	(25)	SENT	Schmidt and Huddle[1977]
6.9	1.00	(58)	(29)	SENT	Schmidt and Huddle[1977]
20.7	1.55	(146)	(73)	SENT	Schmidt and Huddle[1977]
34.5	2.1	(267)	(135)	SENT	Schmidt and Huddle[1977]
48.3	3.05	(530)	(265)	SENT	Schmidt and Huddle[1977]
62.1	4.20	(1005)	(503)	SENT	Schmidt and Huddle[1977]
0	0.66-			IPTWC	Abou-Sayed[1977]
	1.01				
6.0	1.37			IPTWC	Abou-Sayed[1977]
6.3	1.42			IPTWC	Abou-Sayed[1977]
6.9	1.53			IPTWC	Abou-Sayed[1977]
Lueders 1st	0.58	(16)	(8)	HF	Zoback[1978]
	0.37	(6.6)	(3.3)	HF	Zoback[1978]
	(0.65)	(21)	10.5	3PB	Friedman et al.[1972]
	(0.89)	(38)	19	DCB	Perkins and Bartlett[1963]
	(0.84)	(34)	17	DCB	Perkins and Krech[1966]
Austin 1st	(0.44)	(16)	8	DCB	Perkins and Bartlett[1963]
Carthage 1st	(1.65)	(77)	39	DCB	Perkins and Bartlett[1963]
	(1.50)	(70)	35	DCB	Perkins and Krech[1966]
Falerans mcrt	1.01	(13.5)	(6.7)	3PB	Henry et al.[1977]
Carrara mbl	0.64	(6.9)	(3.5)	DT	Atkinson[1979a]
Treuchtlinger mbl	0.99	(16)	(8)	3PB	Winter and Rummel[1979]
	(1.39)	(32)	(16)	3PB	Rummel[1975]
St. Pons mbl					
In foliation	0.70	(10.6)	(5.3)	3PB	Henry et al.[1977]
plane, parallel					
to foliation					
In foliation	1.39	(42)	(21)	3PB	Henry et al.[1977]
plane, normal					
to foliation					
Normal to	1.07	(25)	(12.5)	3PB	Henry et al.[1977]
foliation					
plane, normal					
to foliation					

TABLE 2. (continued)

Material	K_{Ic}, MPa $m^{1/2}$	G_{Ic}, J m^{-2}	Q_f, J m^{-2}	Method	Reference
		Granitic Rocks			
Westerly granite	1.74	(56)	(28)	DT	Atkinson and Rawlings[1981]
	(1.79)	(60)	30	VI	Swain and Lawn[1976]
	0.90	(15)	(7.5)	HF	Zoback[1978]
	2.5	(117)	(58)		Schmidt and Lutz[1979]
Epprechstein Fichtelgebirge granite	1.6	(58)	(29)	3PB	Winter and Rummel[1979]
Chelmsford granite					
Orientation 1	0.64	(12.3)	(6.2)	3PB	Peng and Johnson[1971]
Orientation 2	0.63	(7)	(3.5)	3PB	Peng and Johnson[1971]
Orientation 3	0.59	(6.1)	(3)	3PB	Peng and Johnson[1971]
Carnmellis granite	2.16			DT	Winter[1979]
Pink granite	1.66	(44)	(22)	DT	this study
Bohrung(Werne) granite	0.88-1.04			3PB	Winter[1979]
Urach gneiss	1.47-1.74			3PB	Winter[1979]
Barre granite	(0.74-1.5)	(19-77)	(9.5-39)		Summers et al.[1971]
		Basic Rocks			
Black gabbro	2.88	(82)	(41)	DT	Atkinson and Rawlings[1981]
Academy Black gabbro	0.84			HF	Zoback[1978]
Whin Sill dolerite	3.28			DT	Meredith and Atkinson[1983]
Ralston basalt intrusive	2.58			DT	Meredith and Atkinson[1983]
Preshal More basalt	2.50			SR	Meredith et al.[1981]
Icelandic tholeiite	0.87			SR	Meredith et al.[1981]
		Ultrabasic Rocks			
Dunite	3.74			DT	P.G. Meredith et al. (unpublished manuscript,1983)
Serpentinized dunite	1.39			SR	Meredith et al.[1981]
		Sedimentary Rocks			
Siltstone					
4.4% porosity	1.37			IPTWC	Clifton et al.[1976]
9.5% porosity	1.04			IPTWC	Clifton et al.[1976]
Shale	1.29			IPTWC	Clifton et al.[1976]
0.2% porosity	0.95			IPTWC	Clifton et al.[1976]
3.2% porosity	0.61			IPTWC	Clifton et al.[1976]
6.7% porosity	0.87			IPTWC	Clifton et al.[1976]

TABLE 2. (continued)

Material	K_{Ic}, MPa $m^{1/2}$	G_{Ic}, J m^{-2}	Q_f, J m^{-2}	Method	Reference
Rome basin shale	1.2-1.34				Abou-Sayed et al.[1978]
		Ice			
Temperature, °C					
-16.7	0.05-0.06	0.3-0.4	(0.15-0.2)	THS	Gold[1963]
-12 to -4	0.12-0.15	1.7-2.5	(.85-1.25)	CA	H. W. Lui and L. W. Loop (as reported by Goodman and Tabor [1978])
-12 to -4	0.18-0.22	3.6-5.6	(1.8-2.8)	OGC	
-45 to -4	0.14-0.10	3.0-1.0	(1.5-0.5)	CT	
-13	0.12	1.5	(0.75)	3PB	Goodman and Tabor[1978]
-16	>0.30	>10.0	(>5)	CI	Goodman and Tabor[1978]
-20	0.24	7	(3.5)	CI	Goodman and Tabor[1978]
-20	0.21	5	(2.5)	VI	Goodman and Tabor[1978]
-38	0.07	0.5	(0.25)	VI	Goodman and Tabor[1978]
		Other Materials			
Polycrystalline MgAl spinel	2.0	(8-16)	4-8	3PB	Rice et al.[1978]

All data are for room conditions, unless otherwise indicated. Figures without parentheses are experimental data. Figures in parentheses are calculated from equation (4).

Method: 3PB, three point bend; VI, Vickers indentation; CI, cone indentation; CT, compact tension; DT, double torsion; SR, short rod; OGC, Obreimov-Gilman cleavage; THS, thermal shock; CA, crack arrest; IPTWC, internally pressurized thick-walled cylinder; HF, hydrofracture; DCB, double cantilever beam; SENT, single edge notched tension; WDCB, wedge-loaded DCB.

Material: grnt, granite; sst, sandstone; lst, limestone; qtzt, quartzite; mcrt, micrite; mbl, marble.

When known, crack plane and direction of crack propagation are given.

TABLE 3. Experimental Values of Fracture Mechanics Parameters and Calculated Equivalents for Minerals in Mode I

Material	K_{Ic}, MPa $m^{1/2}$	G_{Ic}, J m^{-2}	Q_f, J m^{-2}	Method	Reference
Quartz, synthetic					
a normal to r	0.85	(6.98)	(3.49)	DT	Atkinson[1979b]
a normal to z	1.00	(9.65)	(4.83)	DT	Atkinson[1979b]
$(10\bar{1}1)$	(1.74)	(23)	11.5	3PB	Hartley and Wilshaw[1973]
$(\bar{1}011)$	(0.33)	(0.82)	0.41	OGC	Brace and Walsh[1962]
(1011)	(0.28)	(1.00)	0.50	OGC	Brace and Walsh[1962]
$(10\bar{1}0)$	(0.41)	(2.06)	1.03	OGC	Brace and Walsh[1962]
$(10\bar{1}0)$	(0.46)	(2.68)	1.34	VI	Atkinson and Avdis[1980]
(0001)	(0.31)	(0.92)	0.46	VI	Atkinson and Avdis[1980]
Quartz, natural Brazilian					
Normal to <c>	2.4			3PB	Norton and Atkinson[1981]

TABLE 3. (continued)

Material	K_{Ic}, MPa $m^{1/2}$	G_{Ic}, J m^{-2}	Q_f, J m^{-2}	Method	Reference
Feldspar					
Orthoclase					
(001)	(1.30)	(15.54)	7.77	OGC	Brace and Walsh[1962]
Microcline					
(001)	(0.39)	(6.40)	3.20	VI	Atkinson and Avdis[1980]
(010)	(0.39)	(10.50)	5.25	VI	Atkinson and Avdis[1980]
(110)	(0.31)	(7.48)	3.74	VI	Atkinson and Avdis[1980]
Olivine					
$Fo_{88}Fa_{12}$					
(010)	(0.59)	(1.96)	0.98	VI	Swain and Atkinson[1978]
(001)	(0.73)	(2.52)	1.26	VI	Swain and Atkinson[1978]
Calcite					
$(10\bar{1}1)$, 77°K	(0.16)	(0.46)	0.23	OGC	Gilman[1960]
$(10\bar{1}1)$	(0.20)	(0.70)	0.35	OGC	Santhanam and Gupta[1968]
$(10\bar{1}1)$	(0.19)	(0.54)	0.27	VI	Atkinson and Avdis[1980]
Muscovite					
(001)	(0.21)	(0.75)	0.38	OGC	Obreimoff[1930]
Spinel					
(111)	(1.27)	(1.88)	0.94	VI	Atkinson and Avdis[1980]
	(1.90)	(9.70)	(4.85)	3PB	Stewart and Bradt[1980]
1000°C	1.8			3PB	Stewart and Bradt[1980]
1500°C	2.75			3PB	Stewart and Bradt[1980]
(100)	1.18	(7.14)	(3.57)	3PB	Stewart and Bradt[1980]
900°C	1.0			3PB	Stewart and Bradt[1980]
1500°C	3.45			3PB	Stewart and Bradt[1980]
(110)	1.54	(8.14)	(4.07)	3PB	Stewart and Bradt[1980]
900°C	1.5			3PB	Stewart and Bradt[1980]
1500°C	3.1			3PB	Stewart and Bradt[1980]
Halite					
(100)	(0.18)	(0.66)	0.33	OGC	Gilman[1959]
Fluorite					
(111)	(0.30)	(0.90)	0.45	OGC	Gilman[1960]
Periclase					
(100)	(0.79)	(2.4)	1.2	OGC	Gilman[1960]
Galena					
(001)	(0.176)	(0.8)	0.4	VI	Atkinson and Avdis[1980]
Pyrite					
(001)	(0.96)	(4.8)	2.4	VI	Atkinson and Avdis[1980]
Diamond					
(111)	(3.4)	(10.1)	5.5	HI	Field and Freeman[1981]
(111)	(3.9)	(14.2)	7.1	HI	Field and Freeman[1981]

All data are for room temperature, unless otherwise indicated.
OGC, Obreimov/Gilman cleavage; VI, Vicker's indentation; 3PB, three point bend; DT, double torsion; HI, Hertzian indentation.

TABLE 4. Compilation of Subcritical Crack Growth Parameters for Geological Materials in Mode I

Material	Conditions	Temperature, °C	n	Activation enthalpy, kJ mol^{-1}	Reference
		Quartz			
a plane, normal to r	air, 30%R.H.	20	19.9		Atkinson[1979b]
a plane, normal to z	water	20-80	12	52.5	Atkinson[1979b]
	2N NaOH	20	9.5		Atkinson and Meredith[1981]
	2N HCl	20	19.3		
	air(0.1-0.5 Pa)	20	41		Meredith and Atkinson[1982]
				67	
		200	40		
				99	
		300	46		
	water vapor (300 Pa)	20	18		Meredith and Atkinson[1982]
				65	
		100	16		
				73	
		200	12		
	water vapor (2000 Pa)	20	13.3		Meredith and Atkinson[1982]
5° to [0$\bar{1}$10]	water vapor	90-240		108	Martin[1972]
	water vapor	20-250		63	Martin and Durham[1975]
<c>	water vapor	20-50		46-100	Scholz[1972]
		Quartz Rocks			
Arkansas novaculite	water	20-80	25	70	Atkinson[1980]
Mojave quartzite	water	20	30-38		this study
	water	40	39.1		this study
Tennessee sandstone	air, 30%R.H.	20	17.2		this study
	water	20	13.7		this study
	water	60	26.3		this study
		Calcite Rocks			
Falerans micrite	air	20	130		Henry and Paquet [1976]
	water+$CaCO_3$	20	26		
	water+$CaCO_3$	20-85		63-147	Henry[1978]
St.Pons marble	water+$CaCO_3$	20	26-29		Henry and Paquet [1976]
Solnhofen limestone	carbonated water	20	26-28		this study
Carrara marble	air, 30%R.H.	20	18.8		this study
	water	20	15.3		this study
		Basic Rocks			
Black gabbro	air, 30%R.H.	20	32.1		Atkinson and Rawlings[1981]
	water	20	28.6-36.1		

TABLE 4. (continued)

Material	Conditions	Temperature, °C	n	Activation enthalpy, kJ mol^{-1}	Reference
Murata basalt	air	20	22		Sano and Ogino [1980]
Ralston basalt	air, 28%R.H.	20	44.4		Meredith and Atkinson[1983]
	water	20	23.5		
	air, 40-50%R.H.	20	97		Swanson[this issue]
Kinosaki basalt	air	30	34		Waza et al. [1980]
	water	25	33		
Whin Sill dolerite	air, 30%R.H.	20	31.2		Meredith and Atkinson[1983]
	water	20-80	29	30-46	
Dresser basalt	air, 40-50%R.H.	20	169		Swanson[this issue]
Tholeiitic basalt	air, 40-50%R.H.	20	171		Swanson[this issue]
		Granitic Rocks			
Westerly granite	air, 30%R.H.	20	39.1		Atkinson and Rawlings[1981]
	water	20	34.8		
	air, 30%R.H.	20	35.9		Atkinson et al. [1980]
	water	20	33.7		
	water	20	35.3-45.9		this study
	water	20	53		Swanson[1980]
	toluene, 11.3%R.H.	20	51		Swanson[1980]
	air, 40-50%R.H.	20	69		Swanson[this issue]
	heat treated to (°C), air				
	20	20	39.1		Atkinson et al. [1983]
	100	20	38.5		
	300	20	39.1		
	500	20	38.6		
Oshima granite	air	20	30		Sano and Ogino [1980]
Lac du Bonnet granite	air	20	58.5		Wilkins[1980]
	air	20	55.9		
Yugawara andesite	air	25	31		Waza et al. [1980]
	water	25	26		
		Sedimentary Rocks			
Devonian shale	air, 40-50%R.H.	20	80		Swanson[this issue]

Subcritical crack growth index n. R.H. is relative humidity.

TABLE 5. Calculated Values of Thermodynamic Surface Energy Q_s

Material	Q_s, J m^{-2}	Reference
Quartz		
($1\bar{0}11$)	0.76	Brace and Walsh [1962]
($10\bar{1}1$)	0.45	Brace and Walsh [1962]
($1\bar{1}00$)	0.48	Brace and Walsh [1962]
Orthoclase feldspar		
(001)	0.20	Brace and Walsh [1962]
Olivine		
forsterite		
(010)	0.37	Brace and Walsh [1962]
(010)	8.63	Swain and Atkinson [1978]
(001)	12.06	Swain and Atkinson [1978]
Calcite		
($10\bar{1}0$)	0.38	Gilman [1960]
Aragonite		
(010)	0.27	Brace and Walsh [1962]
Spinel		
(111)	1.78	Brace and Walsh [1962]
Barite		
(001)	0.48	Brace and Walsh [1962]
Fluorite		
(111)	0.54	Gilman [1960]
Halite		
(100)	0.31	Gilman [1959]
(110)	0.35	Gilman [1959]
Hematite		
(0001)	0.98	Brace and Walsh [1962]
Galena		
(100)	0.63	Gilman [1959]
(110)	0.44	Gilman [1959]
Periclase		
(100)	1.31	Gilman [1959]
(110)	2.33	Gilman [1959]
Pyrite		
(100)	1.55	Brace and Walsh [1962]
(100)	3.16	Babič [1981]
(111)	4.73	Babič [1981]
Rutile	1.42	Brace and Walsh [1962]
Sphalerite		
(110)	0.36	Gilman [1959]
Diamond		
(111)	5.3	Field and Freeman [1981]
(110)	6.5	Field and Freeman [1981]
(100)	9.2	Field and Freeman [1981]

The influence of temperature on K_I-v behavior at different water vapor pressures is shown in Figures 3 and 4. Activation enthalpies, as given in Table 4, are rather low when compared with estimates of activation enthalpy for other deformation processes [Atkinson, 1979b; Meredith and Atkinson, 1982]. The influence of temperature on fracture toughness is shown in Figure 5. This trend mirrors the influence of temperature on Hertzian indentation hardness as reported by Swain et al. [1973].

The dependence of crack velocity in quartz on partial pressure of water vapor has been studied by Meredith and Atkinson [1982] and indirectly by Martin [1972], Martin and Durham [1975], and Scholz [1972]. The experimental results to date are not inconsistent with the idea that the order of the chemical reaction involved in the stress corrosion process is 1. These experimental data are also not inconsistent with the general equation (9), commonly used to describe stress corrosion in quartz, but more rigorous experimental verification is required.

The values of n found for quartz in vacuum are much higher (40-46) than those found in water vapor (12-18) (Table 4 and Figures 3 and 4). It is assumed that data in Figure 3 correspond to region 3 type fracture (Figure 1) because the slopes are so steep and because Wiederhorn et al. [1974] interpreted crack growth in glasses under similar vacuum conditions as being due to purely thermally activated crack growth processes. Wiederhorn et al. [1974] found that in vacuum the slopes of stress intensity factor/crack velocity curves were 1.3-8 times that found by Wiederhorn and Bolz [1970] for the same glasses

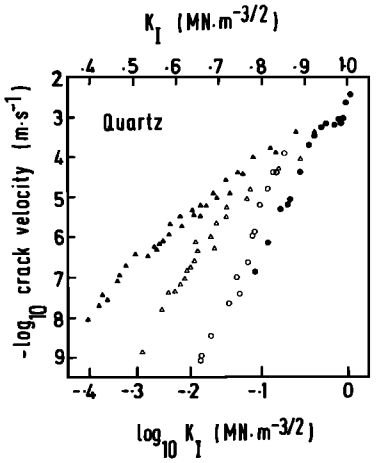


Fig. 2. Subcritical tensile crack growth data for quartz at room temperature (stress intensity factor/crack velocity data). Shows influence of crystallography and "humidity" of environment. Note three-stage behavior shown schematically in Figure 1 [after Bruner, 1980]. Solid and open circles are for crack growth on r in the direction [10$\bar{1}$2] for two specimens of natural quartz at 62.5% relative humidity [Bruner, 1980]. Solid triangles are for crack growth perpendicular to z on a for synthetic quartz in liquid water [Atkinson, 1979b]. Open triangles are for crack growth perpendicular to r on a in synthetic quartz at 68% relative humidity [Atkinson, 1979b].

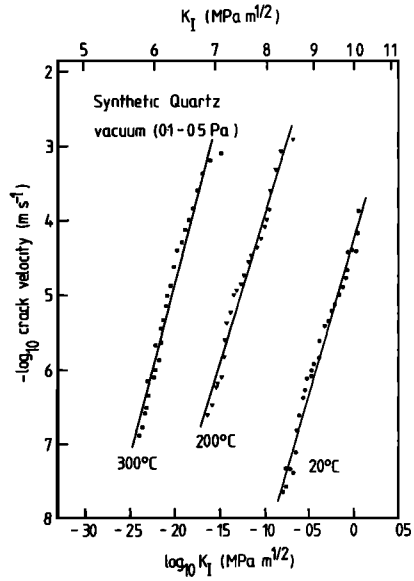


Fig. 3. Influence of temperature on subcritical tensile crack growth in synthetic quartz under vacuum (0.1-0.5 Pa). Crack growth on a plane in a direction normal to z. Stress intensity factor/crack velocity diagram. After Meredith and Atkinson [1982].

in water. The slopes of vacuum data for quartz are about 3.5 times that found for quartz in water. Once the vacuum data are excluded, the range of slopes for stress intensity factor/crack velocity data for quartz cracked on the a plane normal to z is 12-18. There are uncertainties in the data for geological materials when obtained with the relaxation form of the double torsion test (the experimental method used here) that are probably due to the inherent complexity of these materials (see the appendix and Swanson [this issue]). If one considers this and the findings of other workers on single crystals using different methods (e.g., the sapphire data of Wiederhorn [1968] and Figure 17), this spread of results is probably still consistent with the schematic Figure 1. There is apparently a region 2 plateau developed in quartz at 200°C under a water vapor pressure of 300 Pa at a velocity of about 10^{-3} m s^{-1} (Figure 4).

Figure 6 shows the quite substantial influence of pH on crack growth behavior. For a given value of stress intensity factor the crack velocity increases as the hydroxyl concentration is raised. The effect is most marked at low values of stress intensity factor, diminishing as this parameter is raised and at values of stress intensity factor close to K_{Ic} no significant influence of pH on crack propagation rates is observed. This is interpreted by Atkinson and Meredith [1981] in terms of the two-stage model presented in the earlier section on ion exchange. That is, at high crack velocities (stress intensity factors close to the critical value), crack growth rate is controlled by the composition of new crack surfaces, whereas when crack velocity is lowered, the chemistry of the bulk external environment becomes an increasingly important influence on crack velocity [Wiederhorn and Johnson, 1973; Wiederhorn, 1978].

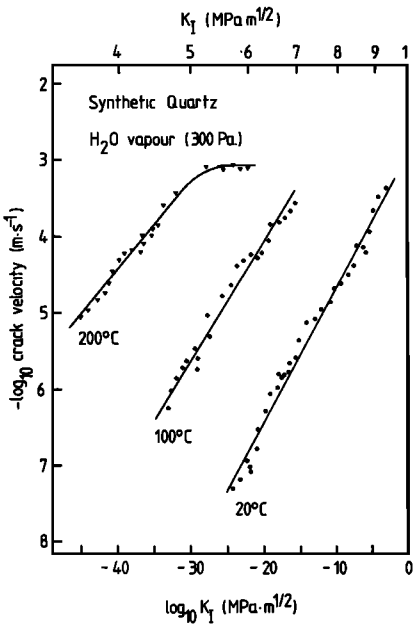


Figure 4. Influence of temperature on subcritical tensile crack growth in synthetic quartz under water vapour pressure of 300 Pa. Other details as for Figure 3. After Meredith and Atkinson [1982].

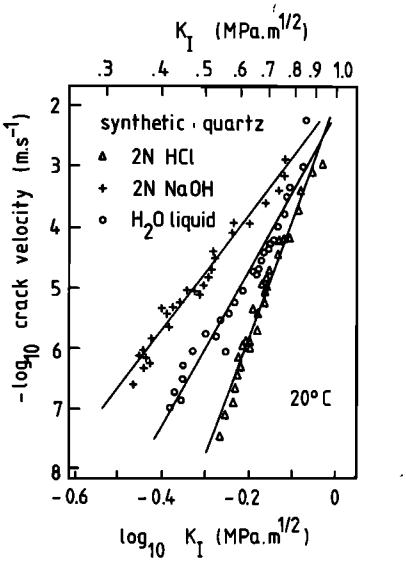


Fig. 6. Influence of pH of aqueous fluid environment on subcritical tensile crack growth in synthetic quartz at room temperature (stress intensity factor/crack velocity diagram). Upper curve (crosses), 2N NaOH; middle curve (circles), liquid water; lower curve (triangles), 2N HCl. Crack growth on a plane normal to z. After Atkinson and Meredith [1981].

Quartzite and Sandstone

Figure 7 shows the influence of a number of variables on subcritical crack growth in quartz rocks. The data for Tennessee sandstone and Mojave quartzite have not previously been published. They were obtained using double torsion specimens tested in the load relaxation mode following procedures described by Atkinson [1979b, 1980].

Apparently trimodal stress intensity factor/crack velocity behavior is found for Mojave quartzite at 20°C in liquid water, but not for any other quartz rock over the range of environmental conditions studied here. Cracks in the very impure Tennessee sandstone apparently support higher values of stress intensity in water at 60°C than they do at 20°C. Uncertainties in these data, however, mean that more work is needed before these features can be viewed as material characteristics and not artifacts of the testing method (see the appendix and discussion below).

The influence of temperature on crack growth has only been studied in a systematic way for Arkansas novaculite [Atkinson, 1980] and the relevant data are given in Table 4. The activation enthalpy for crack growth is similar to that found for single crystals of synthetic quartz.

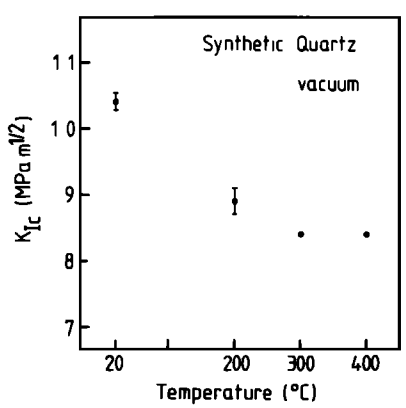


Fig. 5. Influence of temperature on fracture toughness K_{Ic} of synthetic quartz in a vacuum of about 0.1 Pa. Cracked on a plane in direction normal to z. Mode I loading, double torsion tests. After Meredith and Atkinson [1982].

Granitic Rocks

Subcritical crack growth has been studied in several granitic rocks, although Westerly granite remains the most favored material. Figure 8 shows most of the available data for Westerly granite at 20°C. Data for heat-treated Westerly granite tested at 20°C in air [Atkinson et al., 1983] are not included because of the influence of such treatment on the maximum stress intensity that can be supported.

There is a substantial difference between the slope of the K_I-v curves obtained by different workers using nominally the same testing material but slightly different versions of the double torsion testing technique (see the appendix). The data of Atkinson and Rawlings [1981] and Atkinson et al. [1980] show that the slopes of K_I-v curves for different samples of Westerly granite when deformed in the same laboratory by the double torsion method can be reasonably well reproduced. Hence the differences noted above must be due to differences in experimental method and not to variation in starting material. Note

also, however, that there is considerable uncertainty in the placement of stress intensity factor/crack velocity curves on the stress intensity factor axis for specific environmental conditions. This is manifest as an uncertainty in the preexponential constant in equation (5) (see the appendix and later discussion).

It is not possible simply to explain the different results in terms of specimen length (see the appendix) as Swanson [this issue] suggests. Further work is needed to provide an explanation of these differences. This would be especially worthwhile because of the great utility of the double torsion method in studies of rock fracture. In the meantime, however, it should be borne in mind that there are problems in interpreting the results of double torsion tests (see the appendix).

Region 1 behavior is observed in most cases, but in toluene (effective relative humidity is 11.3%) Westerly granite shows a very gradual rounding of the K_I–v curve that suggests the onset of a plateau, or region 2, at a crack velocity of about 10^{-3} m s^{-1} (Figure 8). This gentle change in slope contrasts with the rather more abrupt transition found in silicate glasses and some ceramics. The crack tip in Westerly granite under the test conditions consists of many secondary microcracks with different orientations and different intergranular/intragranular character. The microcracks also pass through or round different host mineral grains. The gradual change in slope can thus be explained as a result of the differential onset of region 2 behavior for different secondary cracks [Swanson, 1980· Atkinson and Rawlings, 1981].

Swanson's [1980] data for Westerly granite in toluene (effective relative humidity is 11.3%)

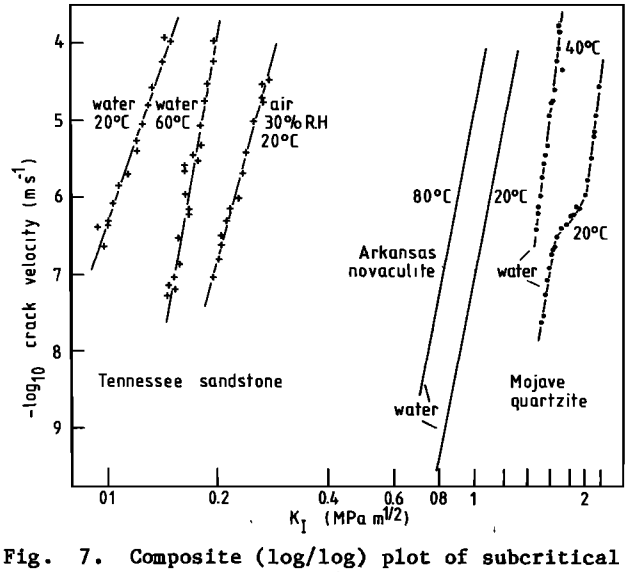


Fig. 7. Composite (log/log) plot of subcritical tensile crack growth data (stress intensity factor versus crack velocity) for several quartz rocks. Shows influence of temperature and water vapor pressure. Crosses, Tennessee sandstone; circles, Mojave quartzite (both from this study); unornamented lines, Arkansas novaculite [from Atkinson, 1980].

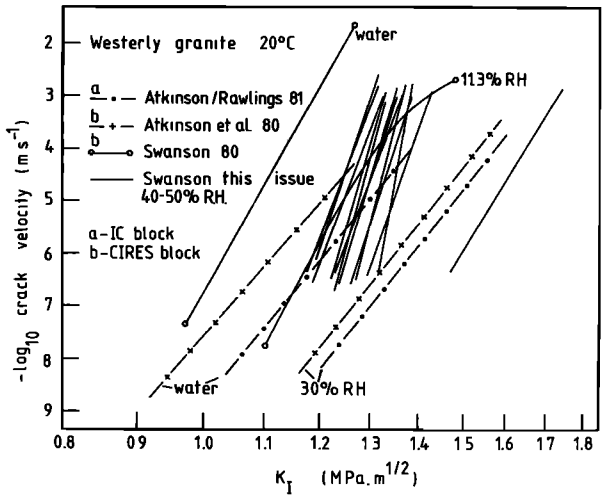


Fig. 8. Synoptic (log/log) plot of subcritical tensile crack growth data for Westerly granite at room temperature. Curve for 11.3% relative humidity obtained in toluene.

apparently lie at a lower stress intensity than most of Swanson's [this issue] data for this material in a more humid environment (40–50% relative humidity) (Figure 8). This is probably an artifact of the double torsion testing method (see discussion in the appendix and by Swanson [this issue]).

In a search for a subcritical crack growth limit in granite, Wilkins [1980] has used a fracture statistics approach to infer crack velocities as slow as 10^{-11} to 10^{-12} m s^{-1}. No such limit was found. These data are compared with most other data obtained at 20°C in the synoptic Figure 9.

Basic Rocks

Figures 10 and 11 and Tables 2 and 4 show much of the relevant data for subcritical crack growth in basic rocks. Other data can be found in the paper by Swanson [this issue]. We still know very little about the subcritical crack growth behavior of basic rocks, however, because of their mineralogical and microstructural complexities.

Comparative studies of subcritical crack growth in Ralston basalt using the same material, but the testing techniques of different laboratories have been made that parallel those previously described for Westerly granite (Figure 11 and Table 4). Once more the difference in the results seems to stem from a difference in testing technique, but the same problems are encountered when trying to explain these differences as found for Westerly granite (see previous section). This point is addressed further in the appendix on the double torsion testing method and in the discussion to this section.

Region 2 behavior is apparently found in tests on Black gabbro in liquid water but not in air of 30% relative humidity over the range of crack velocities studied. The slope of the stress intensity/crack velocity curve at velocities above the apparent region 2 plateau (about 36) is not much different to that at velocities below

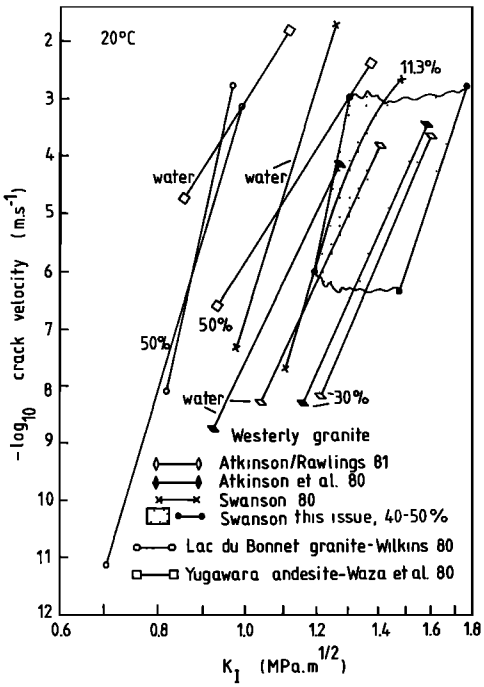


Fig. 9. Synoptic (log/log) plot of subcritical crack growth data for granitic rocks at room temperature. Percentage figures refer to the relative humidity of the environment. Stippled area is the range of data from Swanson [this issue]; mode I loading.

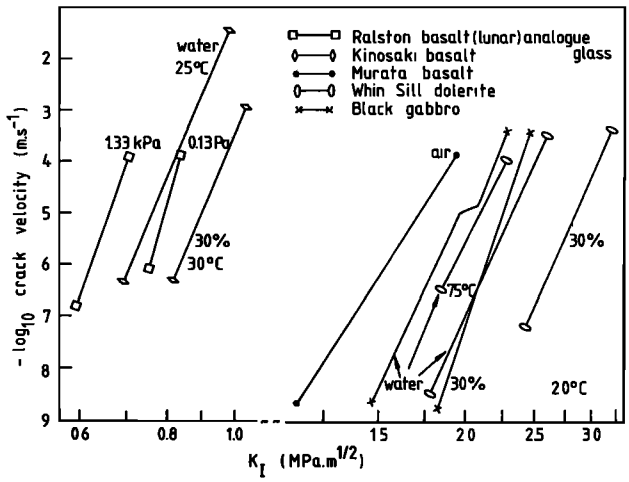


Fig. 10. Synoptic (log/log) plot of subcritical tensile crack growth data for selected basaltic rocks. All data at room temperature unless otherwise indicated. Relative humidity (expressed as percent) or pressure of water vapor (expressed in pascals) of environment given with each curve. Note discontinuous stress intensity factor axis. Sources of data: Ralston basalt and lunar analogue glass [Soga et al., 1979], Kinosaki basalt [Waza et al., 1980], Murata basalt [Sano and Ogino, 1980], Whin Sill dolerite [Meredith and Atkinson, 1983], Black gabbro [Atkinson and Rawlings, 1981].

the apparent region 2 (about 29). This fact, taken with the uncertainty in these data (see the appendix and discussion to this section), means that the plateau may be more apparent than real. This may account for why the region 1 data for Black gabbro in air of 30% relative humidity lie at a higher stress intensity factor than the apparent region 3 for Black gabbro in liquid water. If the plateau is an artifact of the test method, then these data are consistent with the schematic Figure 1.

Comparison of Figures 10 and 11 shows that cracks in a synthetic glass that is a compositional analogue of Ralston basalt support far lower values of stress intensity factor at a given crack velocity than for the crystalline form of Ralston basalt.

Marbles and Limestones

A synoptic diagram of subcritical crack growth data for calcite rocks at room temperature is shown as Figure 12. This diagram contains some previously unpublished data on Carrara marble and Solnhofen limestone. These data were obtained especially for this study using the double torsion testing method in the load relaxation mode following procedures outlined by Atkinson [1979b, 1980].

One may infer from Figure 12 that calcite rocks show rather unusual behavior . Regions 1, 2, and 3 of the schematic stress intensity factor/crack velocity diagram (Figure 1) can be discerned, most clearly in the case of Falerans micrite [Henry, 1978]. In addition to the region 2 plateau in crack velocity, there is apparently a further plateau at much lower crack velocity. On reducing the stress intensity factor below those of the second, low-velocity plateau crack velocity again decreases with decrease in stress intensity factor. The second, low-velocity plateau apparently occurs at a crack velocity of approximately 10^{-6} m s^{-1} in the case of all calcite rocks studied to date. This fact suggests that in the case of calcite rocks the low-velocity plateau may not be an artifact of

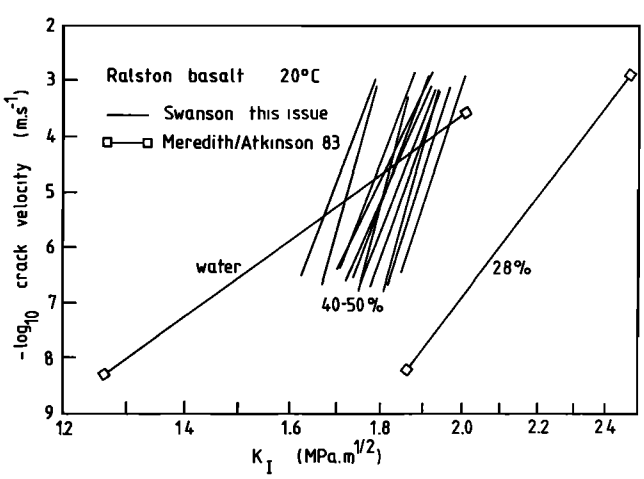


Fig. 11. Synoptic (log/log) plot of subcritical tensile crack growth data for Ralston basalt intrusive at room temperature. All data obtained with the same starting material. Relative humidity of environment (expressed as percent) given with each curve.

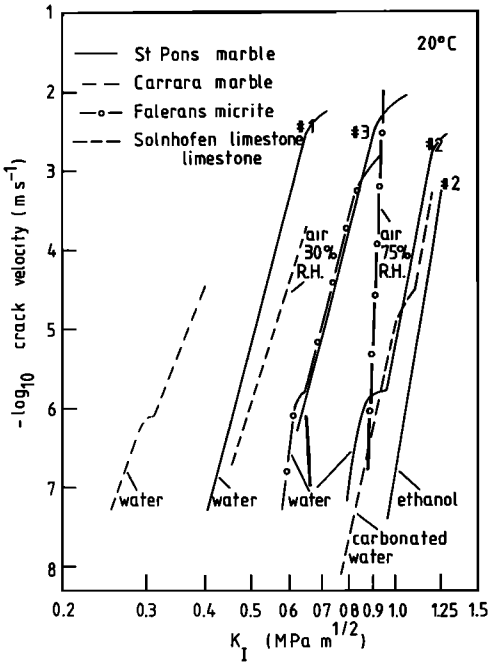


Fig. 12. Synoptic (log/log) plot of subcritical tensile crack growth data for calcite rocks at room temperature. The 1, 2 and 3 refer to three mutually perpendicular directions of crack growth in St. Pons marble. Note bending of curves at crack velocities of about 10^{-2} m s^{-1} and also at about 10^{-6} m s^{-1}. St. Pons marble and Falerans micrite data from Henry [1978]; Carrara marble and Solnhofen limestone data are from this study.

the testing method. If it is a real phenomenon, then it is presumably controlled by the fundamental physicochemical properties of calcite rather than any aggregate properties because it is found at nearly the same crack velocity in calcite rocks of widely differing microstructures. Henry [1978] suggested that the dissolution of calcite may be an important contributor to crack growth in the region of this second, low-velocity plateau.

Using the experimental data on calcite dissolution of Berner and Morse [1974], it is possible to calculate likely rates of crack

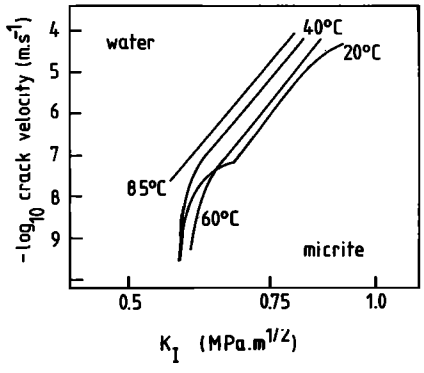


Fig. 13. Influence of temperature of water on subcritical tensile crack growth in Falerans micrite (log/log plot) [after Henry, 1978].

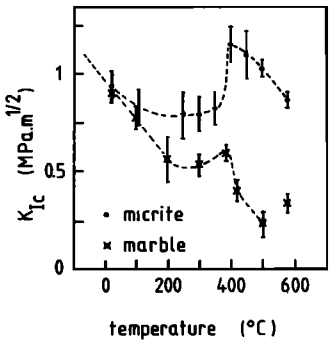


Fig. 14. Influence of temperature on fracture toughness K_{Ic} of Falerans micrite (solid circles) and St. Pons marble (open circles with brands). Mode I loading. After Henry [1978].

growth due solely to dissolution at the crack tip in calcite rocks. At room temperature, crack velocities up to 10^{-9} to 10^{-10} m s^{-1} are possible. These are rather lower than the 10^{-6} m s^{-1} of the low-velocity plateau in Figure 12.

The influence of temperature on crack growth is shown in Figure 13 and in Table 4. For a given crack velocity, cracks in micrite [Henry, 1978] apparently support higher stress intensity factors under water at 60°C than at 40°C. Once more, we must be cautious in interpreting this as a real phenomenon as it may be due to uncertainties in these data obtained by the relaxation form of the double torsion testing method. Henry [1978] has studied the influence of temperature on fracture toughness for two calcite rocks. These data are shown in Figure 14.

In view of our observations on heat-treating rocks [Atkinson et al., 1983] and the extensive electron microscope observations [see Barber and Wenk, 1979] on calcite rocks that show the great ease with which this mineral undergoes dislocation climb and recovery at temperatures well below half the melting temperature, I interpret Figure 14 as follows. The reduction in critical stress intensity factor as temperature is raised to about 200°C largely reflects an increase in thermal microcrack density. At temperatures of 300°-400°C, however, dislocation climb and recovery will become important, even over the relatively short duration of the experiments. Crack tips will thus become blunted, and to maintain the same crack velocity, the stress intensity factor effectively has to be raised. At temperatures above 400°C, thermal microcracks presumably become so abundant that they swamp the influence of the thermally induced crack blunting, and thus critical stress intensity factor falls again. This behavior is probably typical of rocks containing minerals that readily undergo dislocation climb.

Henry [1978] has studied the influence of buffered pH on crack propagation in calcite rocks. Changes in the pH of basic aqueous solutions can produce substantial shifts in stress intensity factor/crack velocity curves (Figure 15). As pH is raised, the second low-velocity plateau of stress intensity factor/crack velocity curves (familiar from Figure 12) is apparently no longer observed over the range of experimental data (Figure 15).

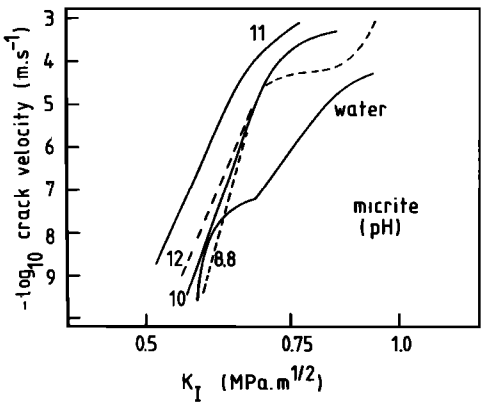


Fig. 15. Influence of buffered pH of aqueous solution on subcritical tensile crack growth in Falerans micrite (log/ log plot). The pH is given by each curve. After Henry [1978].

Other Geological Materials

Subcritical crack growth in shales has been studied by Schmidt [1976b] and Swanson [this issue]. Swanson's data are shown in Figure 16. The values of the slopes of curves in Figure 16 are given in Table 4, and it may be inferred that the curves represent region 1 of the schematic stress intensity factor/crack velocity curve (Figure 1). Note, however, the considerable uncertainty in the placement of the curves on the stress intensity axis (see the appendix and discussion to this section). This uncertainty is due to problems with the relaxation form of double torsion testing referred to earlier [Swanson, this issue]. Schmidt [1976b] conducted time-to-failure tests on an oil shale from Anvil Points, Colorado, in distilled water, air of 7% relative humidity, and dry Argon. He concluded that subcritical crack growth can reduce the time to failure in this oil shale provided that the stress intensity factor was greater than 0.8 K_{Ic}. Because these tests never lasted longer than 123 hours it is possible that subcritical crack growth can occur at yet lower fractions of the critical stress intensity factor but at rates that will result in failure after weeks, months, or years rather than a few hours.

Wiederhorn [1968] obtained stress intensity factor/crack velocity curves for subcritical crack growth of (10$\bar{1}$2) fractures in sapphire when water vapor was present. The experimental data spanned a range of crack velocities from 10^{-4} to 10^{-8} m s^{-1} and showed most of the features of the schematic Figure 1. Even these widely respected data, however, show some scatter in the position of region 1 curves (Figure 17). In several cases the region 1 curves apparently do not lie at lower values of stress intensity as the humidity of the environment increases. At low crack velocities some curves lie at higher stress intensities as the humidity of the environment increases, and this is linked to the slope of region 1 curves apparently not being constant. Lawn [1975] has used these data (obtained by double cantilever beam tests) to illustrate the utility of his atomistic theory of subcritical crack growth in explaining experimental data.

One of the most important features of Wiederhorn's data is that the plateau, or transport-controlled region (region 2) is shifted to substantially higher crack velocities on raising the pressure of water vapor. For example, the plateau occurs at a crack velocity of approximately 7 x 10^{-7} m s^{-1} under a water vapor pressure of 2.4 Pa (relative humidity = 0.003). On raising this pressure to 300 Pa (relative humidity = 0.5) the plateau is shifted to a crack velocity of approximately 7 x 10^{-5} m s^{-1}.

Discussion of Experimental Data

The range of critical stress intensity factors encountered for rocks and quartz is shown as Figure 18. To aid comparison of data, only those obtained with the double torsion method are shown here. In general, for silicate rocks the less quartz there is present and the more ferromagnesian minerals there are then the greater is the value of the critical stress intensity factor. Complicating influences, such as porosity, grain size, etc., can, of course, upset this pattern. Data for other methods show similar trends.

The range of stress intensity factor/crack velocity data for geological materials in liquid water at 20°C is presented in Figure 19 in a format following that adopted by Swanson [this issue]. Again, to aid comparison, only data obtained with the double torsion method are shown, and water was chosen as the test medium as this is a convenient and reproducible reference state. Nevertheless, because of small differences in testing technique and the complications of microstructure this diagram should not be taken as a true ranking of the subcritical crack growth susceptibility of geological materials but rather as a comparison of existing experimental results. As pointed out by Swanson [this issue], it is possible to make one important generalization from diagrams such as Figure 19, however, and that is that the more complex the microstructure, the lower is the susceptibility to subcritical crack growth. Data for an environment of moist air rather than water show a similar trend to

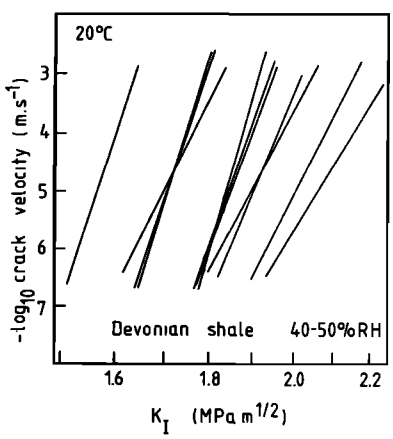


Fig. 16. Subcritical tensile crack growth data (log/log plot) for Devonian shale at room temperature and in air of 40-50% relative humidity [after Swanson, 1982].

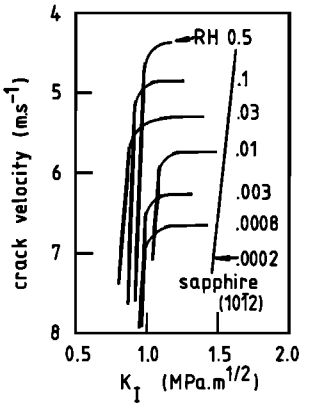


Fig. 17. Subcritical tensile crack growth data (log/linear plot) for sapphire in nitrogen atmospheres of different humidity at room temperature. Note classical three-stage behavior. Data are from Wiederhorn [1968].

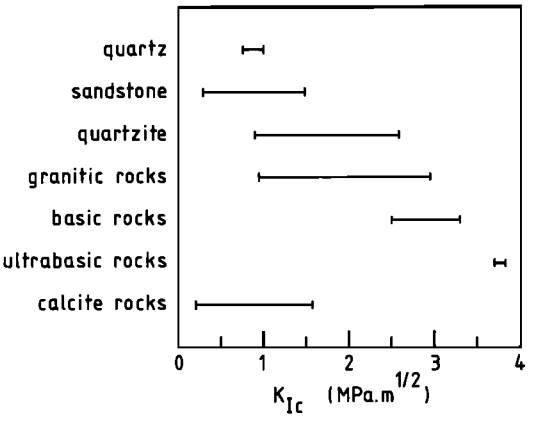


Fig. 18. Variation of fracture toughness K_{Ic} with geological material. All data were obtained with the double torsion testing method at room temperature in air of ambient humidity. Mode I loading.

Figure 19 but with steeper slopes (see also Swanson [this issue]).

To date there has been no experimental verification of the existence of a subcritical crack growth limit in any geological material. It has been inferred by Wilkins [1980] that his experimental studies on granite reached crack velocities as slow as 10^{-11} to 10^{-12} m s^{-1} without a limit being encountered. If a crack growth limit exists, however, its magnitude is of considerable geophysical significance. Attempts to infer the value of the subcritical crack growth limit will be presented in a later section.

Most experimental data on subcritical tensile crack propagation in geological materials appear to be in region 1 or region 3 of the schematic stress intensity factor/crack velocity curve (Figure 1). For tests in water vapor a high crack velocity plateau (region 2 of Figure 1) is apparently sometimes found, for example, quartz (Figures 2 and 4), granite (Figure 8), and dolerite [Meredith, 1983].

There is a range of anomalous features of the experimental data (outlined in the previous sections) that are apparently not consistent with the schematic Figure 1. These include (1) the apparent presence of a high crack velocity plateau region for tests in liquid water on quartzite, calcite rocks, and gabbro, and (2) slopes of region 1 curves for materials in environments of different water vapor pressure that are not always constant.

Most, if not all, of these apparent anomalies can be reconciled with the schematic stress intensity factor/crack velocity diagram (Figure 1) by a consideration of the likely uncertainty in the parameters n and v_o (see the appendix and Swanson [this issue]). It seems that for the relaxation form of the double torsion test the value of n is quite reproducible, but the value of the preexponential constant is not. Uncertainty in the value of v_o is manifest as a lack of precision in locating the curves of Figure 1 on the stress intensity axis. At constant crack velocity a variation in K_I of as much as 20% K_{Ic} can be expected. Moreover, transients in relaxation data in double torsion tests have been reported [Swanson, this issue] and it may well be that these can contribute anomalous plateau regions to stress intensity/crack velocity curves.

The possibility remains, however, that the plateau regions observed for tests in liquid water are real. They have been identified by workers studying ceramics [Evans and Linzer, 1973]. If they are real, then it is not easy to explain their existence in terms of transport-limited crack velocities (as was done for tests in water vapor) because the water should have had constant access to crack tips. Although a complete explanation remains to be provided for this phenomenon in ceramics, it has been suggested that stress corrosion reaction

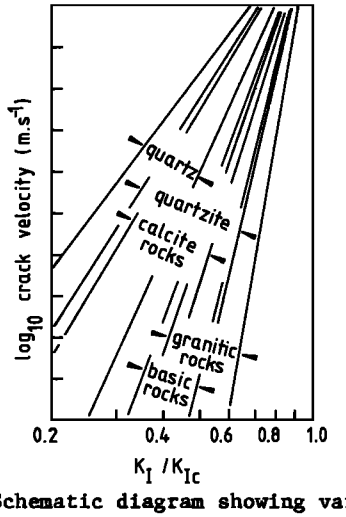


Fig. 19. Schematic diagram showing variation of subcritical tensile crack growth behavior with geological material (log/log plot). Note normalized stress intensity factor axis. Slope of solid lines is the subcritical crack growth index n. All data were obtained with the double torsion testing method at room temperature in liquid water as a convenient reference state. Arrows indicate range of data; maximum and minimum slopes are shown.

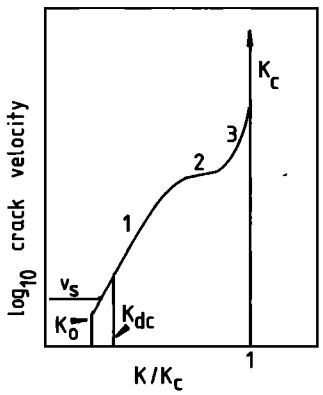


Fig. 20. Schematic diagram showing the influence of stress intensity factor on subcritical crack growth behavior in a rock undergoing stress corrosion. K_c and regions 1, 2, and 3 of stress corrosion behavior are as for Figure 1. K_o is the stress corrosion crack growth limit, v_s is the crack velocity threshold due purely to dissolution at the crack tip, and K_{dc} is the crack growth limit below which bulk diffusion creep dominates deformation behavior. K_{dc} may be located at stress intensities above or below K_o, depending on the temperature. Its location in this figure is chosen for convenience of representation. Note normalized stress intensity factor axis. Temperature, pressure, and activity of stress corrosion agent are all held constant.

products may build up in crack tip regions to create a near-surface barrier through which corrosive water species must be transported to enable stress corrosion reactions to occur [Evans and Linzer, 1973].

Extrapolation of Fracture Data to Crustal Conditions

The environmental conditions that pertain in the earth's crust have been the subject of several reviews of which a recent comprehensive one is by Fyfe et al. [1978]. The reader is referred to this work for detailed discussion of the magnitude of crustal environmental variables. I have outlined in an earlier section some of the key variables that will influence subcritical crack growth. The available experimental evidence on how these variables will influence subcritical crack growth is presented above. For several of the key variables, however, we have little or no relevant information on geological materials. In order to predict fracture behavior in the earth's crust we must extrapolate the limited data base and draw inferences from theory and from related work on ceramics and silicate glasses. That is the purpose of this section. It is intended also to serve as a guide to the planning of future experiments.

The variables whose influence will be examined in the context of crustal deformation are stress intensity factor (especially the nature of the subcritical crack growth limit), temperature, activity of stress corrosion agent, pressure, grain size, microstructure and process zone, and residual strains. In addition, other problems of extrapolating crack growth data to the earth's crust are discussed including the appropriate equation(s) to describe crack growth and the magnitude of the parameters in these equations.

Influence of Stress Intensity Factor

Figure 20 shows a schematic diagram of the influence of stress intensity factor on subcritical crack growth due to stress corrosion at constant temperature, pressure, and activity of the stress corrosion agent. At high stress intensity factors it is similar to Figure 1, the schematic K-v diagram obtained from work on silicate glasses and ceramics, except that the transition from region 1 to region 2 and from region 2 to region 3 for rocks are more rounded than in these other materials. To recap, in region 1 the crack velocity is apparently controlled by the stress corrosion reactions, in region 2 the rate of transport of reactive species to crack tips is believed to control the crack velocity, and in region 3 the crack velocity is controlled by some poorly understood, thermally activated process that is comparatively (but not totally [Freiman, this issue]) insensitive to the chemical character of the environment.

At very slow tectonic rates of strain the low stress intensity factor/low crack velocity region becomes of great interest. Here the K-v diagram becomes more complex than the relatively simple Figure 1. It is assumed that there exists a threshold K_o below which no significant crack extension can occur by stress corrosion. This is a function of the material's fracture properties and its environment. The value of K_o is calculated in a later section, but it will suffice here to note that it will be a small fraction (10-20%) of the critical stress intensity factor.

At stress intensity factors above the stress corrosion crack growth limit, significant crack growth will be suppressed if the stress is high enough for diffusion creep to make a significant contribution to deformation rate and thereby blunt macrocrack tips. This has been observed in ceramics [Karunaratne and Lewis, 1980]. This threshold will be dependent upon the strain rate and the grain size of the rock. It may give rise to a limit to subcritical crack growth at K values above K_o. This is given as K_{dc} in Figure 20.

Another type of low crack velocity behavior may be possible if dissolution of the cracked solid can proceed at a rate comparable to the stress corrosion crack velocity. The high stress concentration at crack tips presumably localizes this behavior [Simmons and Freiman, 1981]. It could lead to a threshold crack velocity v_s, as shown in Figure 20. This behavior has been inferred from experimental data on binary alkali-silicate glasses [Simmons and Freiman, 1981] and calcite rocks (see above).

Influence of Temperature

Figure 21 shows a schematic drawing of the influence of temperature on a K-v diagram. Crack growth in all three regions is thermally

activated and so they are all influenced by temperature changes.

In region 1 except over restricted temperature intervals the detailed physicochemical mechanism of subcritical crack growth and the rock microstructure will not be constant and the slopes of K-v curves will not be parallel. In the Charles-Hillig-Wiederhorn theory [Wiederhorn and Bolz, 1970] the slope of a K-v curve will vary with temperature. Experimental data on crack growth in glass under vacuum (presumably region 3 of Figure 1) also show this behavior [Wiederhorn, 1978].

At high homologous temperatures, crack growth due to diffusion of vacancies to a crack tip may occur. The limited experimental evidence for this process shows that in ceramics the stress intensity exponent n in equation (5) is rather low (n = 10) [Karunaratne and Lewis, 1980], that is, much lower than for stress corrosion in rocks (Table 4).

Krausz [1978] notes that if the mechanism of subcritical crack growth does not change, then K_o is independent of temperature. The threshold of crack growth suppression due to diffusion creep K_{dc}, however, will be shifted to higher values of stress intensity factor as temperature is raised because diffusion creep is faster at higher temperatures. In this context, large-scale diffusion creep is acting to suppress subcritical crack growth by crack blunting rather than the grain scale diffusion creep that can facilitate crack extension and which was discussed earlier.

If there exists a threshold crack velocity due to dissolution in calcite rocks (v_s), it will be shifted to lower crack velocities (v_s^*) as temperature is raised. For quartz rocks the threshold (v_s) will be shifted to higher crack velocities (v_s') as temperature is raised. This behavior results because these minerals show retrograde and normal temperature dependence of solubility, respectively.

One may infer from the empirical experimental data of Meredith and Atkinson [1982] and Meredith [1983] that the absolute value of K_c is likely to drop as temperature is raised if deformation is brittle. This is probably due to the development of thermal microcracks which when linked can facilitate crack extension. If the temperature rise is small, however, and the thermal microcracks remain isolated, then they may tend to blunt rather than assist macrocrack growth and this may lead to a small increase in the maximum stress intensity that a crack can sustain. If significant crack tip microplasticity can occur then K_c may rise as temperature increases due to plastic deformation blunting crack tips thus requiring the value of stress intensity factor to be raised in order to achieve the same point of instability.

Munz et al. [1981] show that if the microscopic mechanism of crack growth changes at high homologous temperatures, as in certain ceramics, then crack branching and secondary cracking may increase (process zone size increases) and the crack path may change from being a mixture of intergranular and transgranular cracking to become dominated by intergranular processes.

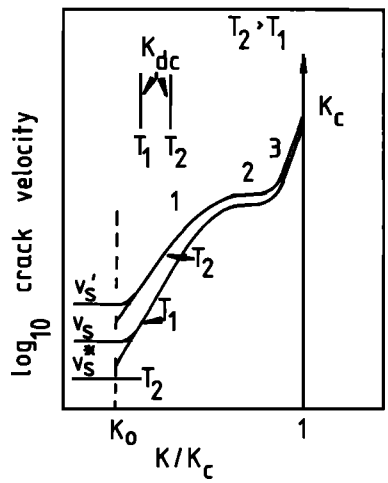


Fig. 21. Schematic diagram showing influence of temperature on subcritical crack growth in a rock undergoing stress corrosion. K_c and regions 1, 2, and 3 are as for Figure 1. Temperature $T_2 > T_1$. K_o is stress corrosion crack growth limit. Note that stress intensity factor axis is normalized with respect to fracture toughness at the relevant temperature. The v_s is the threshold crack velocity due purely to dissolution at the crack tip at temperature T_1. The v_s' is the threshold crack velocity due to dissolution at crack tip at temperature T_2 in the case of normal solubility (e.g. quartz). The v_s^* is the threshold crack velocity due to dissolution at temperature T_2 in the case of retrograde solubility (e.g. calcite).

Influence of Activity of Stress Corrosion Agent

The chemical activity of the environmental agent is perhaps the key variable in controlling environmentally dependent, chemically enhanced subcritical crack growth [Freiman, this issue]. There is some controversy over whether ionized water or molecular water is the key agent for subcritical cracking in silicate glasses, quartz, and rocks. To determine which is the important agent is of considerable geophysical interest for the following reasons.

From what we know of the variation in temperature T, total or overburden pressure P_t, and pore fluid pressure P_f in the earth's crust and experimental data on the activity of water ($a[H_2O]$) and ionized water ($a[H^+]$, $a[OH^-]$), two forms of behavior are possible. Note that the activity of agent i (a_i) is given by the expression f_i/f_{io}, where f_i is the fugacity of agent i and f_{io} is its fugacity in some standard state. Note also that the fugacity of a component is equal to its partial pressure in an ideal mixture.

First, it can be seen from Figure 22 that for molecular water $a[H_2O]$ increases with P_f, T, and depth in the crust. Therefore, if activity of molecular water is the key agent in stress corrosion cracking, then an increase in crack growth rate at constant stress intensity factor would be expected as depth in the earth's crust was increased. Over most of the earth's crust, however, the fugacity of molecular water changes

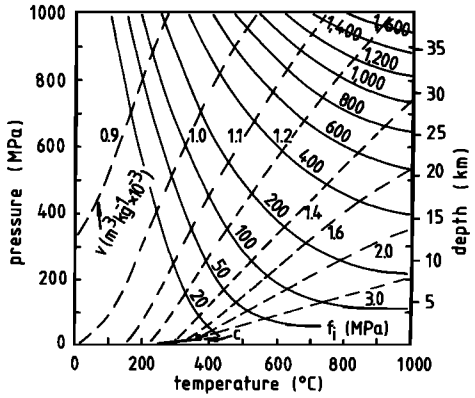


Fig. 22. Specific volumes V (in m^3 kg^{-1})(inverse of density), dashed lines, and fugacities f_i (in megapascals), solid lines, of water as a function of temperature and pressure. Depth values assume a constant crustal density of 2600 kg m^{-3}, geothermal gradient 20°C km^{-1}. Critical point is denoted by letter c. After Fyfe et al. [1978].

by only about one and a half orders of magnitude. For comparative purposes, the data of Figures 3 and 4 suggest that increasing the partial pressure of water by this amount is only likely to increase crack velocity in quartz by about the same order of magnitude.

Second, water ionizes according to

$$H_2O \leftrightarrows H^+ + OH^- \qquad (14)$$

and its ion product K_w is given by

$$K_w = a_{OH^-} \cdot a_{H^+} \qquad (15)$$

It will be seen from Figure 23 that the degree of ionization of water increases with water pressure and increases with depth (when pore fluid pressure is close to lithostatic pressure). The degree of ionization will undergo a slight increase as temperature is raised to 250°C, but it will actually decrease as temperature is raised further. Substantial changes are possible in the crustal values of K_w. At surface conditions, for example, $K_w = 10^{-14}$, but at the base of the crust it is close to 10^{-8}.

Because water becomes a stronger acid than most common acids at depth it will displace them from their salts. For example,

$$NaCl + H_2O \leftrightarrows NaOH + HCl \qquad (16)$$

Thus neutral or acid conditions should be rare in high-temperature crustal environments [Fyfe et al., 1958].

If ionized water, especially hydroxyl, is the key environmental agent in stress corrosion cracking of rocks, then large variations in crack velocity for a fixed stress intensity factor would be expected, mirroring the large variation in $a[OH^-]$ expected in the earth's crust.

The fact that $a[H_2O]$ increases with temperature but that $a[OH^-]$ decreases with increasing temperature above 250°C suggests a means whereby the key environmental agent could be isolated by experiment.

Figure 24 shows a schematic diagram of the expected K-v behavior when the activity of the agent is changed at constant temperature and pressure. Region 3 shows only a very weak dependence on chemical weakening processes and so is depicted as showing no change as activity of the agent is varied. The higher the activity of the agent, the greater is the crack velocity due to stress corrosion at constant stress intensity factor.

The subcritical crack growth limit may be displaced to lower values of K as activity of stress corrosion agent increases. The limited experimental data for copper and steel [Stanzl and Tschegg, 1981] suggest that a 30% reduction in K_0 can be achieved when a relatively chemically inert environment is replaced with 3.5% sodium chloride solution. No relevant experimental data are available for rocks.

Influence of Pressure

Anderson and Grew [1977] point out that we must be careful to distinguish three effects of pressure on subcritical stress corrosion crack growth. These are the effects of pressure on (1) the stress tensor of the solid undergoing crack growth, (2) the mechanical properties of the fluid in the crack, and (3) the chemical reactions taking place between the fluid in the crack and the solid material at the crack tip. If the kinetics of the stress corrosion reaction or the chemical potential of environmental agent and reactants at the crack tip are influenced by pressure, then one would expect to see a corresponding influence on crack velocity [Anderson and Grew, 1977]. This will be true for subcritical crack growth by diffusional processes as well as by stress corrosion.

It is also necessary to consider how pressure

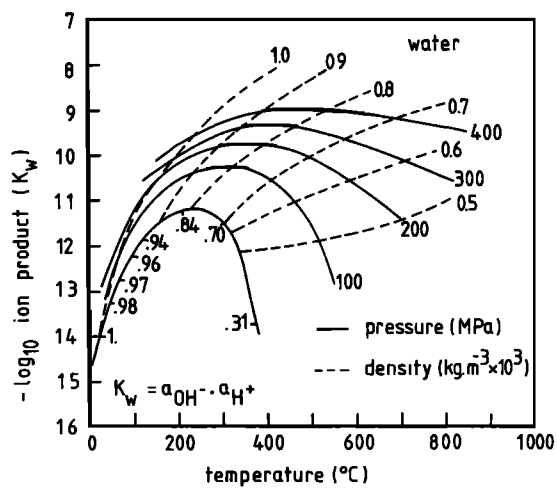


Fig. 23. Log of ion product of water as a function of temperature, density, and pressure [after Fyfe et al., 1978]. Lowest solid curve shows point values of density of liquid water along liquid plus vapor saturation curve. To infer behavior at depth in earth's crust to compare with Figure 24 data, assume a geothermal gradient of 20°C km^{-1} and a geobaric gradient of 26 MPa km^{-1}.

will influence the development of microcracks and the process zone at the macrocrack tip. This will mainly influence effect 2 above.

It is extremely difficult to separate out the different aspects of the influence of pressure on subcritical crack growth behavior. For this reason a schematic diagram showing the influence of pressure on subcritical crack growth is not included. As Anderson and Grew [1977] conclude, pressure could well increase crack velocity in one situation and decrease it in another. The nature of the influence of pressure on subcritical crack growth will of course depend on the micromechanisms of crack growth.

The fracture toughness of triaxially loaded Indiana limestone has been studied by Schmidt and Huddle [1977] and by Abou-Sayed [1977]. Abou-Sayed found that the apparent fracture toughness of internally pressurized, thick-walled cylinders increased by as much as 80% when the effective confining pressure was raised to about 6.5 MPa. Schmidt and Huddle, on the other hand, found that the apparent fracture toughness of single edge notched specimens increased by only 6% when the effective confining pressure was raised to 7 MPa. Both Abou-Sayed [1977] and Schmidt and Huddle [1977] agree on the fracture toughness of Indiana limestone at room pressure.

The general influence of pressure on stress intensity factor will now be considered. Even for tensile fracture one may reasonably assume that a critical microcrack density is required in the process zone (nonlinear microcracked zone) at macrocrack tips before linking is possible to allow the extension of the macrocrack. If we can assume that these process zone microcracks are dominantly tensile in character (see Ouchterlony [1980], and note also that microcracking during macroscopically compressive failure is also dominantly tensile [Tapponnier and Brace, 1976; Holcomb and Stevens, 1980]), then the application of hydrostatic confining pressure will reduce any applied tensile stress across crack tips and hence retard the ability of the rock to develop a critical microcrack density. Thus a greater remote macroscopic tensile stress is required to propagate a macrocrack.

Kaieda and Oguchi [1981] and Kaieda and Pae [1982] have examined the problem of hydrostatic pressure on crack tip extension conditions. We apply their work here to a geophysical context. For surface cracks or blunt cracks, when the pore fluid pressure has access to crack tips and $P_f = P_t$ (i.e., $P_e = 0$) and σ is the applied macroscopic tensile stress, then

$$K_I = Y(\pi c)^{1/2}\sigma \qquad (17)$$

For conditions where the pore fluid has access to crack tips but P_f is less than P_t (i.e., P_e is greater than 0), then

$$K_I = Y(\pi c)^{1/2}\,(\sigma - P_e) \qquad (18)$$

The stress intensity factor decreases as effective pressure increases because the effective tensile stress at crack tips is being reduced. Macroscopically, however, to maintain the same rate of crack extension, it is necessary to increase the remote tensile stress in order to achieve the same crack tip stress intensity

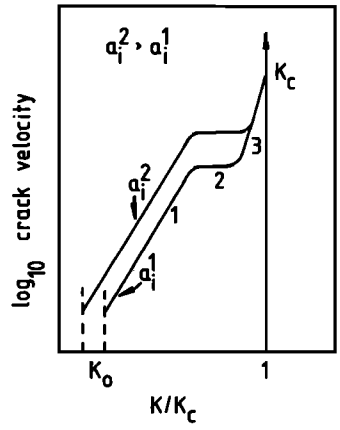


Fig. 24. Schematic diagram showing influence of activity of environmental agent a_i on subcritical crack growth behavior in a rock undergoing stress corrosion. Threshold K_{dc} and v_s omitted for simplicity. Note normalized stress intensity factor axis.

factor as previously. Thus fracture toughness calculated using the applied tensile stress (apparent fracture toughness) will increase as hydrostatic pressure is raised, but the fracture toughness calculated using the differential tensile stress (true fracture toughness) is independent of pressure [Kaieda and Oguchi, 1981]. It is important to make this distinction between apparent and true fracture toughness because the critical stress intensity factor is a crack tip criterion for failure. We can now consider aspects of pressure on crack growth other than its influence on crack tip stress intensity conditions.

The details of the influence of pressure on region 1 of the schematic K-v curve (Figure 1) are not at all clear. There are at present no direct data on the influence of pressure on stress corrosion crack growth rates in rocks, although Kranz [1980] has studied the influence of pressure on time to failure of triaxially compressed granite (see below). There is even some controversy and confusion over the sign of the activation volume for stress corrosion of quartz glass [Ito and Tomozawa, 1981].

Wiederhorn and Johnson [1971] found in their study of static fatigue of glasses in water that changing the water pressure from atmospheric to 600–700 MPa had no effect on static fatigue properties and, by implication, there should be little effect of pressure on stress corrosion crack velocity.

In region 2 (Figure 1), increasing effective pressure should decrease the crack wall spacing and hence retard stress corrosion crack propagation by inhibiting the access of reactive species to crack tip sites of stress corrosion. Raising effective pressure should therefore shift the plateau region 2 (where the crack outruns the environment) to a lower crack velocity. An increase in the viscosity of fluid in cracks due to application of pressure will have the same effect.

Region 3 (Figure 1) is only slightly sensitive to chemical environmental effects and so one may infer that it will be influenced by pressure

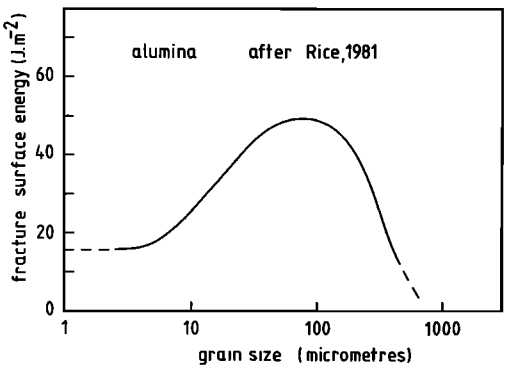


Fig. 25. Schematic diagram showing influence of grain size on fracture surface energy Q_f of alumina [after Rice, 1981]. This behavior is expected to be typical of rocks comprised of non-cubic minerals.

largely to the extent that pressure influences the stress tensor.

Kranz [1980] has found that increasing the effective confining pressure increases static fatigue time to failure of triaxially compressed granite and, by inference, inhibits subcritical crack growth. Three sources of this effect have been suggested by Kranz: a large negative activation volume, crack closure under pressure, and an inhibiting of crack-linking processes (although the last two effects may be difficult to distinguish in practice). The activation volume is not known, but even if it is of the correct sign it would be surprising if it were large enough to account for the magnitude of Kranz's results. Crack-linking effects will, presumably, be most important in explaining the results of Kranz, especially when taken with crack closure effects that will control the placement of region 2. The higher the effective confining pressure the greater was the total microcrack volume prior to failure.

Dehart and Liebowitz [1968] suggest that the absolute value of K_o for some metals undergoing stress corrosion may be increased by about 25% through the application of approximately 70 MPa pressure. This result is in fair agreement with the theoretical prediction of Gerberich [1974]. No relevant data are available for rocks.

Influence of Residual Strains

It is well known that rocks often contain residual strains of palaeotectonic origin [Price, 1966; Friedman, 1972; Holzhausen and Johnson, 1979; Hoskins and Russell, 1981]. Although these stresses are not often considered in rock mechanics investigations, they can be important in rock deformation because they will modify the local stress conditions in an aggregate. Several mechanisms can contribute to the development of residual strains in rocks, including the following:

1. Crystal grains in an aggregate are cemented while under load and then later unloaded, leading to compressive stresses in the grains and tensile stresses in the cement matrix.
2. Volume changes in mineral grains, such as quartz and feldspar, during the cooling of igneous bodies.
3. Differential plastic deformation during tectonic or other forms of loading.

These residual strains may be relieved to some extent by fractures created in response to internal or external stresses, but they are nevertheless common in crustal rocks, especially granites.

The primary effect of macroscopic residual stress on subcritical crack growth is to cause the macrocrack front to be in a state of tension or compression before any external load is applied. By thus modifying the crack tip stress tensor, tensile residual stress augments tensile crack growth and compressive residual stress retards tensile crack growth.

Unpublished work has been done in our laboratory to determine the fracture toughness of a granite containing residual stresses. The short rod testing technique was used as this allows the magnitude of the effect of residual stress on fracture toughness to be estimated [Barker, 1981]. Our work suggests that there may be changes in fracture toughness caused by macroscopic residual stress effects of at least 20%. Changes in K_{Io} of 30-50% have been found in metals due to macroscopic residual stress [Vosikovsky et al., 1980].

Microscopic residual stress distributed on the scale of the grain size may influence the slope of K-v curves because the locked-in stress pockets will act as either obstacles or aids to crack growth. In this context they should act much as microstructural features would (i.e., cracks, pores, heterogeneous mineralogy). The scale of the heterogeneity in residual stress relative to the crack length and specimen size is important here. The microstructural aspects of subcritical crack growth are discussed in the next section.

Influence of Microstructural Features

It is generally observed that for noncubic brittle ceramics, K_c, G_c, or Q_f increases with increase in grain size up to the point at which some critical grain size is reached, whereupon further increase in grain size results in a reversal of this trend (see Figure 25) [Rice et al., 1980, 1981; Mussler et al., 1982; Rice and Freiman, 1981; Singh et al., 1979]. The details of experimental data on the grain size dependence of fracture toughness are strongly dependent, however, upon the test method employed. Studies of alumina and other ceramics have shown that for all but the finest-grained materials (less than 5 μm), fracture toughness increases with crack extension (R curve behavior). The test method dependence of results can be explained therefore by the fact that some test methods allow more crack extension than others before catastrophic crack propagation [Mussler et al., 1982].

The initial increase in fracture energy (or toughness) with increase in grain size of ceramics is normally associated with a shift from intergranular fracture in very fine-grained materials to transgranular fracture in coarser-grained materials [Rice et al., 1980; Mussler et al., 1982]. The nature of the transition depends on the textural properties of the solid and on the cleavage properties of the grains. For alumina, at least, there is a rather

sharp transition from intergranular to transgranular fracture with increasing grain size above 5 μm. Before the fracture properties of a polycrystalline material are characterized by polycrystalline values it is necessary that the flaw size/grain size ratio be greater than 1-6 [Rice et al., 1980].

The maximum in the fracture energy/grain size diagram (Figure 25) is attributed by Rice and Freiman [1981] to the following. Noncubic ceramics (and most rock-forming minerals are also noncubic) show thermal expansion anisotropy. Cooling from the fabrication temperature produces stresses between grains of different orientation due to this thermal expansion anisotropy. These stresses augment the influence of the applied stress in the formation of intercrystalline microcracks about the advancing macrocrack. This microcracking absorbs energy that would otherwise be available to propagate the macrocrack. As the grain size increases above some threshold for the onset of this behavior, the extent of microcracking will increase, absorbing more energy, so that the fracture toughness will also increase. When microcracking becomes too extensive, however, macrocrack growth can take place by linking of microcracks, thereby reducing fracture toughness. These opposing trends lead to a maximum in fracture toughness with grain size.

The observation that the proportion of intergranular fracture in alumina decreases with increase in grain size [Mussler et al., 1982] is apparently not consistent with the implications of the Rice and Freiman model. Wu et al. [1978] found for alumina and other ceramics that there was an increasing occurrence of crack branching on increasing the grain size. This has led Mussler et al. [1982] to suggest that increasing fracture toughness with grain size may be due to low-velocity crack branching. Grain boundary cracking may help to initiate this phenomenon.

For geological materials the many complicating factors, such as porosity, preexisting microcrack density, fabric, and mineralogical variation, preclude the sort of study possible on well-controlled ceramic microstructures. What data there are, however, suggest that there is an increase in fracture toughness with increase in grain size. Very coarse grained specimens are generally not studied in the laboratory because of the need for very large specimens to ensure a valid test. Thus it has not so far been possible to test whether the trend of Figure 25 describes the behavior of rocks. Fine-grained basic rocks (Ralston basalt, K_c = 2.58 MPa $m^{1/2}$, and Preshal More basalt, K_c = 2.5 MPa $m^{1/2}$) have lower K_c values than coarser-grained ones (Black gabbro, K_c = 2.88 MPa $m^{1/2}$, and Whin Sill dolerite, K_c = 3.28 MPa $m^{1/2}$) when tested under identical conditions [Meredith and Atkinson, 1983]. Highly pure quartz rocks show a similar trend. Arkansas novaculite (grain size is less than 10 μm, K_c = 1.34 MPa $m^{1/2}$) has a lower K_c than coarser-grained Mojave quartzite (grain size is 200 μm, K_c = 2.1 MPa $m^{1/2}$) when tested under identical conditions (see Table 2).

There are many conflicting and complicating factors, but generally, the fracture energy of grain boundaries is less than the fracture energy of cleavage because of impurities, porosity, second phase particles, and imperfect bonding at grain boundaries, all of which tend to weaken them. Also, the grain boundary fracture energy for large grains is generally less than the grain boundary fracture energy for small grains because the concentration of grain boundary defects is greatest for large grain sizes.

Subcritical cracking, especially at low crack velocities, is associated with intergranular crack propagation. At fast crack velocities, under stress intensity factors close to the critical value, transgranular cracking becomes more important and eventually predominates behavior at very fast crack velocities [Swanson, 1980· Atkinson and Rawlings, 1981 Mecholsky, 1981].

Microstructural variables also exert an influence on crack propagation through their influence on the process zone or region of microcracking or nonlinear behavior at macrocrack tips. Increasing the density of microcracks in a solid can result in a reduction of $K_c(G_c)$ and an increase in the susceptibility to subcritical crack growth. It can also lead, however, to a reversal of this trend (i.e., an increase in $K_c(G_c)$) if at high microcrack densities the ratio of the process zone width to radius of microcracks is lower than 1 [Kreher and Pompe, 1981]. This condition is only likely to be relatively easily attained in materials with laminated structures, such as shales. For most others the matrix may become unstable before a sufficiently high microcrack density can be reached.

With a sufficiently high density of preexisting microcracks, macrocrack extension changes from growth of the macrocrack plus subsidiary further microcracking around the macrocrack tip to propagation of the macrocrack by linking and coalescence of the preexisting microcracks [Wu et al., 1981; Swanson, this issue]. The latter mechanism of crack extension is more energy efficient [Swanson, this issue]. From studies of ceramics, Wu et al. [1978] show that the first type of macrocrack extension is accompanied by microcrack formation in the process zone and crack branching and wandering on the scale of several grains. The second type of macrocrack extension involving microcrack linking gives far less crack branching and wandering restricted in scale to close to the dimensions of a single grain.

Swanson [this issue] has suggested that a high resistance to subcritical crack growth (high value of n in equation (5)) is associated with multiple crack formation and branching. When a crack surmounts a barrier, the apparent macrocrack velocity is less than if the crack/barrier interaction had not occurred. The more crack/barrier interactions the more tortuous is the crack path and the hence the greater is the resistance to subcritical crack growth (higher values of n). The greater the degree of crack wandering due to structural inhomogeneity, residual strains, microcracks, etc., the greater will be the resistance to subcritical crack growth.

Subcritical Crack Growth Limit

In a geophysical context the low-velocity crack growth limit is perhaps the most important

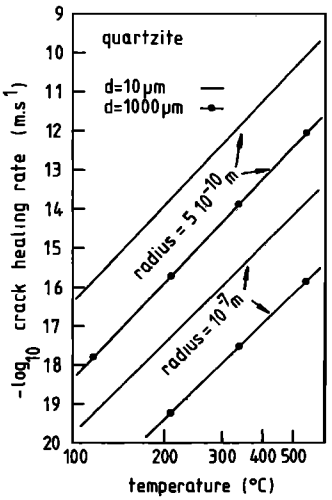


Fig. 26. Crack healing rate as a function of temperature for dry quartzite. Grain size d = 10 or 1000 μm. Two crack tip radii are considered as shown.

parameter describing subcritical crack growth [Das and Scholz, 1981]. Unfortunately, it is extremely hard to measure this parameter in laboratory experiments. Several studies of rocks have been made at crack velocities as low as 10^{-9} m s^{-1} and K_I values < 0.5 K_{Ic} without encountering a crack growth limit (see earlier sections). Moreover, Wilkins [1980] has inferred that his tests on granite reached velocities as slow as 10^{-12} m s^{-1}, again without encountering a crack growth limit. In this section some ways of estimating the value of the subcritical crack growth limit are explored.

A first approach to this problem is to assess the possible rates of crack healing. In crack healing there are several potential mass transport paths, as in crack extension by diffusion processes. For reasons outlined in the section on subcritical crack growth mechanisms we consider here only the grain boundary diffusion path.

Some experimental studies [Bandyopadhyay and Roberts, 1976] have verified that sintering of cracklike pores by grain boundary diffusion of vacancies away from crack tips can be described by the equations of Dutton [1974]. The rate of crack healing (-dc/dt) is given by

$$-dc/dt = [D_b w/4r_c d][1- \exp(-U_a N/RT)] \quad (19)$$

where r_c is the crack tip radius, d is the grain size, N is Avogadro's number, D_b is the grain boundary diffusion coefficient, w is the grain boundary width, R and T are as previously defined, and U_a is the chemical potential of atoms at the crack tip. The chemical potential of atoms at crack tips in the absence of stress is given by [Dutton, 1976]

$$U_a = Q_b V_a / r_c \quad (20)$$

where Q_b is the specific grain boundary surface energy and V_a is the atomic volume. If the crack tip is stressed, then for long thin cracks,

$$U_a = (V_a/r_c)[B\sigma^2 c - Q_b] \quad (21)$$

where $B = \pi(1-p^2)/2E$ and σ is the applied stress. As stress enters equation (21) as a squared term for long thin cracks, then its effect is independent of sign, and both tensile and compressive stresses will reduce the driving force for the shrinking of cracks and the rounding of crack tips. For small circular cracks, compressive stress will assist crack sintering [Dutton, 1976].

We consider here crack healing in the absence of stress. Calculated crack healing rates for quartzite as a function of temperature using equation (19) are given in Figure 26. The chosen values of the variables in equation (19) are as follows: $V_a = 1.9 \times 10^{-29}$ m^3, $w = 10^{-7}$ m, $r_c = 5 \times 10^{-10}$ m or 10^{-7} m, $Q_b = 1$ J m^{-2}, d = 10 or 1000 μm, D_b is given by 2.7×10^{-21} m^2 s^{-1} (200°C), 9.7×10^{-20} m^2 s^{-1} (300°C), 1.2×10^{-18} m^2 s^{-1} (400°C) [Dennis and Atkinson, 1984; and Dennis, this issue].

The crack healing process occurs in two stages. First, long cracks are pinched off and healed to give porelike cracks. Second, these pores undergo further healing and shrink in diameter. Two distinct ranges of crack tip radius are therefore appropriate. Crack tip radius for long thin cracks is taken to be of the order of the lattice spacing and is 5×10^{-10} m [Wiederhorn, 1969; Spetzler et al., 1981; Stevens and Dutton, 1971]. For porelike cracks the radius of the tip could be several orders of magnitude higher than the lattice spacing [Bandyopadhyay and Roberts, 1976], and a value of 10^{-7} m is chosen for convenience. For sharp cracks at temperatures above 300°C, healing rates are quite rapid, but they decline sharply as cracks become more porelike (Figure 26).

We have considered above the case of dry quartz. If water is present at grain boundaries, as will be common in the crust, then the rate of crack healing is expected to be faster as the absorbed water layer and pore water provide faster diffusion paths. The value of appropriate diffusion coefficients is a matter of speculation, however, because appropriate experimental data are unavailable. Hence detailed calculations of this case are not included here. Crude sums indicate that the presence of water may increase the rate of crack healing by three orders of magnitude. Experimental data on crack healing are presented by Smith and Evans [this issue].

The subcritical crack growth limit depends on which crack growth mechanisms operate. When subcritical crack growth involves mass transport processes, the crack growth limit can be defined as the point at which the net flux rate of atoms at the crack tip is zero. For subcritical crack growth involving stress corrosion the crack growth limit can be defined as the point at which the stress concentration due to sharpening cracks is constant [Hillig and Charles, 1965; Shetty et al., 1979], that is, when the semimajor (c) and the semiminor (z) axes of the elliptical crack are given by c/z = const.

We can calculate the subcritical crack growth limit by adopting the Charles-Hillig formulation for stress corrosion crack growth, which can be expressed as [Hillig and Charles, 1965]

$$v = v_o \exp[g\sigma^* + e\sigma^{*2} - (h/r_L)] \quad (22)$$

TABLE 6. Calculated Values of the Subcritical Crack Growth Limits for Quartz

Mechanism	Temperature, °C	Mode I Stress Intensity Factor at Growth Limit, MPa $m^{1/2}$
Stress corrosion		
Case A	20	0.2
	200	0.19
Case B	20	0.39
	200	0.46
Grain boundary diffusion		
Case A	20	0.39
	200	0.22
Case B	20	0.38
	200	0.38

K_{Ic} for quartz is approximately 1 MPa $m^{1/2}$

where v_o is the corrosion rate on an unstressed planar surface, σ^* is the local concentrated stress at any point on a crack surface, r_L is the local curvature of the crack surface, and g, e, and h are factors describing enhancement of corrosion reactions by stress concentration, a strain energy term, and an interfacial energy term (concerning the interfacial energy between solid and the reaction products), respectively. The interfacial energy term h represents the change in free energy associated with the concave surfaces of the crack tip. The interfacial energy term tends to oppose the action of the applied stress and tends to direct the corrosion reactions to blunt the crack tip. The strain energy term is an attempt to represent the change in free energy per mole removed because of the change in total potential energy of the system.

It is necessary to consider two cases to simplify calculations. For case A, where high-order stress terms in equation (22) are negligible, the crack growth limit is given by [Shetty et al., 1979]

$$K_o = [(QV_m)/2br_c] + [(RT)/b] \ln(c/z) \quad (23)$$

where Q is the interfacial energy, V_m is the molar volume, r_c is the crack tip curvature, and b is given by $v^c = v_o \exp(bK_I/RT)$. Here b is the index of stress corrosion susceptibility in the empirical Wiederhorn and Bolz [1970] formulation. If the strain energy term in equation (22) is dominant, case B, then the crack growth limit is given by (following the method of Shetty et al. [1979])

$$K_o = [(QE\pi/2) + (ERT/V_m)(\pi r_c) \ln(c/z)]^{1/2} \quad (24)$$

where E is Young's modulus.

The values chosen for parameters in equations (23) and (24) are as follows. For quartz, V_m = 22.7 x 10^{-6} m^3 mol^{-1}, r_c = 5 x 10^{-10} m, b = 0.058 $m^{2.5}$ mol^{-1} (20°C) and 0.104 $m^{2.5}$ mol^{-1} (200°C) obtained by replotting the data of Atkinson [1979b] in accord with equation (7), c/z=100, and Q = 0.1 J m^{-2}. The value chosen for Q at the low crack velocities and wet environments of a stress corrosion crack growth limit is 1/10 of that in dry conditions when fast fracture occurs and in the diffusion-limited crack growth limit considered below. This is because tests on glasses [Wilkins and Dutton, 1976] suggest that the effective surface energy at the stress corrosion crack limit is 1/10-1/50 of the surface energy for fast fracture. This may be due largely to the fact that at these low crack velocities there is sufficient time for the corrosive environment to maintain contact with the crack tip surface such that crack tip surface energy is reduced by chemical absorption and reaction. Other possible mechanisms are considered by Wilkins and Dutton [1976].

Using the above data, K_o for stress corrosion in quartz is calculated for cases A and B and presented in Table 6. As K_I values below the K_o of case B have been shown experimentally to result in significant crack growth rates in quartz [Atkinson, 1979b; Meredith and Atkinson, 1982], the case where the strain energy term is significant may not be important in practical examples.

For subcritical crack growth involving diffusional mass transport we can adopt a similar course to the above and identify a case A (high-order stress terms negligible) and a case B (strain energy term dominant) and determine crack growth limits after Shetty et al.'s [1979] modification of the theory of Wilkins and Dutton [1976]. For case A,

$$K_o = (QV_m)(2br_c) \quad (25)$$

For case B,

$$K_o = (QE\pi/2)^{1/2} \quad (26)$$

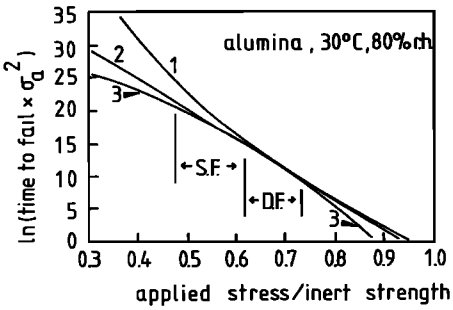


Fig. 27. Influence of the form of equation used to describe subcritical tensile crack growth in alumina on strength/lifetime predictions. After Jakus at al. [1981]. S.F. is range of static fatigue, and D.F. is range of dynamic fatigue experimental data. Curve 1, Charles [1958] equation (27). Curve 2, Wiederhorn and Bolz [1970] equation (28). Curve 3, Lawn [1975] equation (29). Data plotted as ln(time to failure times square of remote applied stress) versus ratio of applied stress to strength in absence of subcritical crack growth (inert strength).

Calculated values for K_o (mass transport) are also presented for quartz in Table 6. For the range of parameters considered here, K_o (mass transport) is always higher than K_o (stress corrosion), and so, in principle, the latter will be the lowest K_I value that can be attained. Note that this conclusion depends on the range of environmental parameters studied and that K_o (mass transport) is not the same as K_{dc} in Figure 20. The latter parameter arises from large-scale bulk diffusion creep and is located at a stress intensity factor that may be above or below K_o (stress corrosion) depending on the temperature.

Some Further Problems of Extrapolation

In addition to all the previously discussed problems associated with extrapolating laboratory results on subcritical crack growth to geophysical conditions, two further ones deserve mention.

The first problem relates to the determination of an appropriate equation to describe subcritical crack growth behavior. Figure 27 shows the calculation of a function of time to failure for alumina in terms of the ratio of applied stress/inert strength (i.e., in absence of subcritical crack growth) after Jakus et al. [1981]. Similar behavior is expected for rocks. The prediction of three subcritical crack growth equations are plotted. These are

$$v = A_1(K_I/K_{Ic})^{n_1} \quad \text{curve 1} \qquad (27)$$

$$v = A_2 \exp[n_2(K_I/K_{Ic})] \quad \text{curve 2} \qquad (28)$$

$$v = A_3 \exp[n_3(K_I/K_{Ic})^2] \quad \text{curve 3} \qquad (29)$$

where A_i and n_i are constants.

Curve 1 (equation (27)) is the Charles [1958, equation (5)] equation. Curve 2 (equation (28)) is the Wiederhorn and Bolz [1970, equation (7)] equation. Curve 3 (equation (29)) is the Lawn [1975] equation. All of these are popular ways of describing subcritical crack growth data.

The problem is that because of the high value of n_i and the inability of the activation enthalpy to distinguish between different competing mechanisms it is virtually impossible to distinguish between the predictions of the three equations over laboratory ranges of static fatigue data (S.F. in Figure 27) and dynamic fatigue data (D.F. in Figure 27). Thus any of the three equations is a good enough fit for laboratory purposes. Unfortunately, in geophysical problems we need to extrapolate beyond the bounds of laboratory testing to low applied stresses and long periods of time to failure. As shown in Figure 27, however, under these conditions the predictions of the three equations diverge substantially, and it becomes of considerable importance to distinguish between the potential equations describing subcritical crack growth.

A second problem is that even assuming we can safely adopt a particular equation to describe subcritical crack growth, for example, the Charles [1958] equation, there is a discrepancy between estimates of the crack growth parameters n_i and A_i for some rocks (see Figures 8 and 11) which also hampers the extrapolation of laboratory data to geophysical environments.

Crack velocity/stress intensity factor data can be used to calculate failure stress σ_f through an equation due to Evans and Johnson [1975],

$$\sigma_f = [2\dot{\sigma}(n+1)/A_1 Y^n (n-2)c_i^{(n-2)/2}]^{1/(n+1)} \qquad (30)$$

where c_i is the initial flaw size, n is n_1 of equation (27), Y is a geometrical constant, and $\dot{\sigma}$ is the stress rate which is given by $\dot{\sigma} = E\dot{e}$, where $\dot{e}$ is the strain rate and E is Young's modulus. Time to failure under constant stress intensity factor can be estimated from

$$t_f = 2/\sigma_a^2 Y^2 \int_{K_{Ii}}^{K_{Ic}} (K_I/v)dK_I \qquad (31)$$

where K_{Ii} is the stress intensity factor at the initial flaw size and σ_a is the applied tensile stress.

Uncertainties in the estimation of parameters n_1 and A_1 in the studies of basic and granitic rocks reported earlier and in the appendix will markedly influence the estimation of time to failure after long times and failure stress at low stress rates. One must conclude that at present such extrapolations are not justified. Also of importance is whether under crustal conditions rocks have subcritical crack growth behavior akin to the results for Westerly granite or whether they are more like the data for Arkansas novaculite. The novaculite shows a much greater susceptibility to subcritical crack growth than the granite. Das and Scholz [1981] have attempted to estimate the subcritical crack growth behavior of crustal rock during delayed multiple event earthquakes. It may be inferred from their data that during tectonic deformation,

crustal rocks show more susceptibility to subcritical crack growth than laboratory results at ambient conditions on Westerly granite.

Summary and Conclusions

An extensive review has been made of the currently available experimental data on subcritical crack growth in geological materials. The available data have been extrapolated to conditions of interest in the earth's crust. A discussion is given of the problems involved in such extrapolations. For variables whose influence on subcritical crack growth in rocks is not well known, schematics of probable trends are presented based upon experimental work on ceramics and on theoretical perspectives. Some of the main conclusions are as follows.

1. There are several possible mechanisms of subcritical crack growth relevant to crustal processes. These include stress corrosion, diffusional mass transport, microplasticity, dissolution, and ion exchange. Theoretical calculations have been made on dissolution and diffusion-controlled subcritical crack growth rates. Dissolution could produce crack growth rates in calcite rocks at room temperature of about 10^{-10} m s^{-1} under tension. Grain boundary diffusion in quartzite could lead to crack growth under tension at velocities of about 10^{-11} to 10^{-13} m s^{-1} at 400°C as grain size changes from 10 to 1000 μm. These are not insignificant crack velocities on a geophysical time scale. The overwhelming body of evidence, however, suggests that at temperatures below 300° C, at least, stress corrosion is the main mechanism of subcritical crack growth over all crack velocities studied in experiments.

2. Microstructural features, such as grain size, porosity, mineralogy, and the nature of the grain boundaries, probably exert a key influence on subcritical crack growth behavior in rocks by controlling the nature of the macrocrack path and the degree of subsidiary microcracking and crack branching.

For silicate rocks the less quartz present and the more ferromagnesian minerals there are, then, in general, the greater is the fracture toughness. Porosity, grain size, and other microstructural features can upset this trend. Coarser-grained rocks generally have higher values of fracture toughness than finer-grained mineralogical equivalents, but data for ceramics and granite suggest that this trend may be reversed at very coarse grain sizes.

The more complex the microstructure the lower is the susceptibility to subcritical crack growth, that is, the subcritical crack growth index is higher.

3. At low crack velocities, subcritical crack growth curves may be terminated by regions controlled by dissolution or general diffusion creep before a stress corrosion crack growth limit is attained. At temperatures of 20°- 200°C the subcritical crack growth limit in quartz due to stress corrosion is calculated to be much lower than that due to diffusional processes and falls at around 0.2 of the critical stress intensity factor.

4. It is not clear whether stress corrosion crack growth in silicates is due to ionized or molecular water. This is of considerable geophysical significance because the potential range of activities of molecular water in the earth's crust is much smaller than that of ionic water species. Above a temperature of about 250°C the activity of hydroxyl ions decreases, whereas that of molecular water continues to increase. This suggests a way of distinguishing between the action of the two forms of water species.

5. Residual strains of palaeotectonic origin can influence both fracture toughness and the details of subcritical crack growth in rocks by influencing the stress conditions at crack tips (macroscopic residual strains) and by acting as microstructural features (microscopic residual strains).

6. With a data base limited to crack velocities that are attained in most laboratory experiments it is impossible to distinguish between the several mathematical models of subcritical crack growth. The predictions of these models diverge significantly at geophysically interesting slow crack velocities, and so this dilemma poses an important limitation on extrapolation of laboratory data on subcritical crack growth to tectonic deformation.

7. Theoretical calculations indicate that geophysically significant rates of crack healing in dry quartzite will be attained once temperatures exceed 300°C. During healing the crack will change its aspect ratio to resemble a pore, and there is a concomitant reduction in the rate of crack healing by several orders of magnitude. To reestablish the earlier rates of crack healing, much higher temperatures would be required.

8. The influence of pressure on subcritical crack growth in rocks is hard to predict. Application of pressure may enhance crack growth in some environments and retard it in others. Pressure can influence (1) the stress tensor of the solid undergoing crack growth, (2) the mechanical properties of any fluid in the crack, (3) the development of microcracks and the process zone at macrocrack tips, and (4) the chemical reactions between any fluid in the crack and the solid material at the crack tip (chemical potentials and reaction rates).

Appendix: The Double Torsion Testing Method

To date, the double torsion testing method [Evans, 1972; Williams and Evans, 1973] has been chosen for virtually all fracture mechanics studies of subcritical crack growth in geological materials. A schematic drawing of a double torsion specimen is shown in Figure A1. In the context of this review it is appropriate to discuss briefly some of the limitations to this technique and and what variables influence the quality of data.

For well-developed cracks such that (L-W)>c>W, where L is the length of the specimen, W is its width, and c is the crack length, the stress intensity factor is given by

$$K_I = PW_m[(3(1+p))/Wt^3t_n]^{1/2} \qquad (A1)$$

where P is the load, p is Poisson's ratio, and the other terms are as defined in Figure A1.

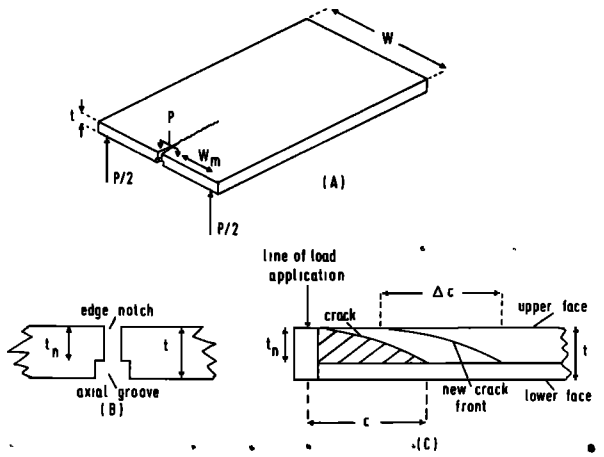


Fig. A1. Schematic drawing of double torsion specimen: (a) general view, (b) axial cross section, (c) longitudinal cross section.

The crack velocity is usually determined by the constant displacement (or load relaxation) method. In this case the crack velocity is given by

$$v = -\phi c_f P_f (1/P^2)(dP/dt) \qquad (A2)$$

where dP/dt is the rate of load relaxation, ϕ is a geometrical constant that takes into account the shape of the crack front and c_f,P_f are measures of the crack length and load, respectively, taken simultaneously at the end of an experiment. Crack velocity can also be determined by constant displacement rate or constant load testing.

The popularity of the load relaxation (constant displacement) method lies mainly in that a very wide range of crack velocities can be explored in one test on a single specimen. The lowest crack velocity that can be attained, however, depends on the stability of the total measuring system. One problem with using the double torsion method in this mode is the relaxation of the loading system. In testing rocks it is usual to employ a very stiff testing machine and cycle a much higher load on the test jig than will be employed during testing (before the test specimen is inserted) until no significant machine relaxation occurs at the load levels to be used in testing.

The principal assumptions inherent in equations (A1) and (A2) are set out clearly by P. L. Swanson (unpublished manuscript 1982): (1) there are no frictional constraints along the sides of the torsion arms (sides of the crack), (2) elastic strain energy providing the driving force for crack extension is derived only from strained torsion arms with negligable deformation ahead of the crack tip, (3) the elastic constants of the sample are independent of test environment, and (4) the force relaxation results only from crack propagation with minimal additions from plastic flow and time-dependent distortion of the loading apparatus.

There has been considerable recent discussion in the literature regarding the constraints on the validity of the double torsion method [Fuller, 1979; Pletka et al., 1979; Pabst and Weick, 1981; Atkinson, 1979a; Annis and Cargill, 1978; Swanson, 1981, this issue; Shetty and Virkar, 1978; Trantina, 1977; Dalgleish et al., 1980; Champomier, 1979; Raju, 1981; Govila, 1980; Meredith et al., 1981]. These studies have improved our understanding of the validity of the test as defined in the original works of Evans [1972], Williams and Evans [1973], and Outwater et al. [1974]. Various factors have been identified as possible sources of errors and uncertainties in double torsion data, and the assumptions of the method have been found not to be satisfied in certain specimen geometries [Swanson, 1981, and unpublished manuscript, 1982]. When used for fracture toughness testing, however, the double torsion method compares well with other techniques, such as the short rod [Meredith et al., 1981].

At least two major problems exist. First, Swanson [this issue, and unpublished manuscript, 1982] has shown that for a given specimen geometry values of n (equation (5)) are repeatable to a degree (see also later in this appendix and Table A1) but that there exist rather large differences in the mean values of n determined for the same materials by different laboratories, for example, contrast data in Figures 8 and 11 and Table 4. These differences cannot be attributed to variations in starting material. Second, as shown by Swanson [this issue], the placement of stress intensity/crack velocity curves on the stress intensity axis is not very repeatable using the load relaxation method (see also Figures 16 and A2). This manifests itself as a poor precision in the preexponential constant in equation (5).

In trying to account for these problems we will address two variables of particular concern, the length of the specimen and the geometry of the axial guide groove.

Pabst and Weick [1981] have shown that short double torsion specimens give the largest values of n. On the other hand, Swanson [this issue] suggests that it is the existence of a narrow range of crack lengths for which the stress intensity factor is a constant at constant load in specimens used by Atkinson and Rawlings [1981] and by other investigators that accounts for the difference between his results and those obtained by these other workers. Values of the subcritical crack growth index n are reduced when a crack propagates into an area of constant stress intensity factor from outside that region [Shetty and Virkar, 1978] or by extraneous apparatus relaxation [Pletka et al., 1979]. Extraneous relaxations were eliminated in both the experiments of Swanson [this issue] and those of Atkinson and Rawlings [1981] and Meredith and Atkinson [1983]. We are inclined to discount the explanation of Swanson [this issue] for the difference between his results and those of Atkinson and Rawlings [1981] and Meredith and Atkinson [1983] for the following reasons.

1. Great care was taken in all tests to ensure that the constant stress intensity factor conditions were satisfied.

2. We have recently been able to reproduce our results using specimens with much longer L:W ratios than in our original studies that are closer to the ratios used by Swanson (see Figure A2 and Table A1). There is also no systematic trend toward greater n values for longer initial

TABLE A1. Values of Subcritical Crack Growth Index n for Westerly granite in Liquid Water at Room Temperature

n	L:W	Reference
39.0	5	test L6/1 Figure A2, this study
45.9	5	test L6/2 Figure A2, this study
36.0	5	test L6/3 Figure A2, this study
35.3	2.7	test S1/1 Figure A2, this study
34.8	2-2.5	Atkinson and Rawlings [1981]
33.7	2-2.5	Atkinson et al. [1980]

All data are for double torsion tests in load relaxation mode. L:W is the length:width ratio of specimens. Mean n = 37.5; standard deviation = 4.5. Data for test L6 are for multiple relaxations on the same specimen. Data are presented in the table in order of increasing crack length.

crack lengths in multiple load relaxation experiments on the same specimen.

The geometry of the guide groove may be a key factor. Wider grooves tend to reduce and narrow grooves tend to increase the likelihood of interactions between crack tips and guide groove wall. The guide groove geometry influences the stress field across the groove so that the crack can be trapped or deflected transiently on traversing the specimen, leading to a scatter in the data [Pletka et al., 1979; Pabst and Weick, 1981; Annis and Cargill, 1978]. Other details of the side groove geometry may be important. Pabst and Weick [1981] show that square-formed guide groove edges may produce higher local stress intensities that accelerate crack growth.

The elimination of the guide groove altogether produces a more curved crack front than that developed when a guide groove is present [Pabst and Weick, 1981] and also reduces the scatter in the data [Pabst and Weick, 1981; Annis and Cargill, 1978; P. L. Swanson, unpublished manuscript, 1982]. It is impractical, however, to eliminate side grooves in such heterogeneous materials as rocks. Using several different guide groove shapes and dimensions did not, however, significantly reduce the scatter in the data of P. L. Swanson's (unpublished manuscript, 1982) experiments on rocks.

One can envisage two extreme cases of crack propagation behavior in double torsion tests. In the first, the guide groove can be made narrow so that the macrocrack is constrained to follow a relatively straight path. In the second, a wider groove can be made so that the crack is presented with a relatively large cross section of rock, and it can choose which is the lowest energy path, the intragranular or intergranular character of which is controlled more by the properties of the rock than by the geometry of the guide groove. A useful clue to the guide groove constraint on crack propagation should be the guide groove area to (grain size)^2 ratio. However, attempts made by the author to correlate this parameter with n values has produced no significant trend.

One must conclude that there is no simple explanation for the different results obtained for the same rocks using slightly different double torsion geometries. More work is needed on this problem.

The data of Figure A2 and Table A1 when combined with those of Atkinson and colleagues in Figure 8 and Table 4 can be used to estimate the uncertainty in our stress intensity/crack velocity measurements for Westerly granite in liquid water using the relaxation form of the double torsion test. These must be considered best estimates for rocks because Westerly granite is more homogeneous, even textured, and fine grained than several crustal rocks that we have also tested in the laboratory. In addition, the liquid water environment is much more easily reproducible than the humid, but unsaturated environments often used in experiments. The mean value of n in these experiments is 37.5, with a

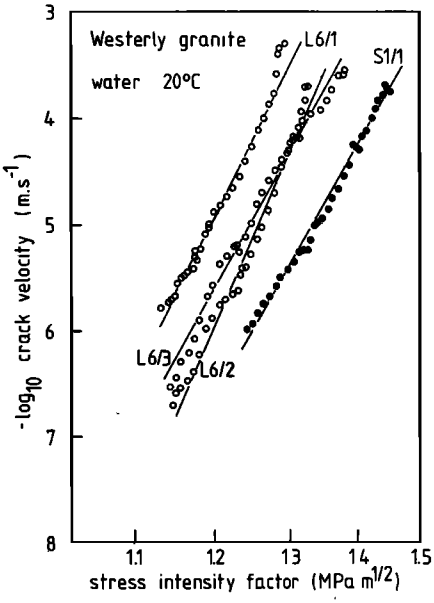


Fig. A2. Subcritical crack growth data for Westerly granite (log/log plot) at room temperature in liquid water, to show repeatability of experimental data using the load relaxation form of the double torsion testing method. Tests L6/1 to L6/3 are repeat relaxations on the same specimen. Test S1/1 is for a single relaxation on a shorter specimen. For details of specimen dimensions see Table A1. Data are from this study.

standard deviation of 4.5. The standard deviation expressed as a percent of n is 12%. The uncertainty in n as determined here is comparable to that found for Westerly granite and a range of other rocks by P. L. Swanson (unpublished manuscript, 1982). The uncertainty in the preexponential constant in equation (5) can be estimated by examining the uncertainty in the placement of stress intensity factor/crack velocity curves on the stress intensity axis. For our data on Westerly granite in liquid water at room temperature (Figures A2 and 8) the uncertainty in placement of the curves on the stress intensity axis is approximately 15%, that is, 10% of the critical stress intensity factor. Swanson [this issue, and unpublished manuscript, 1982] finds that an uncertainty of up to 20% K_{Ic} is to be expected for some rock types (see Figure 16). Even greater uncertainty (20-30% K_{Ic}) in the placement of curves might be inferred from some data on heat-treated Westerly granite in air [MacDonald, 1980], but these data are not included here because of other reservations about the experiments.

In the absence of an international standard and reproducible form of the double torsion test method, a useful test program may be run by using the relaxation form of the double torsion test to obtain n and either the constant load or constant displacement rate form to locate the stress intensity/crack velocity curve on the stress intensity axis [Swanson, this issue].

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